The Limiting Distributions of Chi-square Statistic in High-Dimensional Contingency Tables

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Abstract

We provide necessary and sufficient conditions for the chi-squared and normal approximations of Pearson's chi-squared statistic for the test of independence in two-way contingency tables whose dimensions may diverge. A cross-sample version of Pearson's chi-squared statistic is shown to be more robust than the classic version. A degrees-of-freedom adjusted chi-squared approximation, matching the test statistic in both mean and variance, improves the accuracy and allows a continuous transition from the standardized chi-squared approximation to the normal approximation throughout the low- and high-dimensional regimes. Compared with traditional guidelines for the applicability of the chi-squared approximation, the necessary and sufficient conditions relax the sample size requirement and allow empty cells to dominate the table, provided that a diverging number of cells with more than a single count are expected in any deterministic fractional subset of cells. Necessary and sufficient conditions are provided as well for the chi-squared and normal approximations of Pearson's chi-squared, the log likelihood ratio and Hellinger statistics for the test of goodness-of-fit in one-way contingency tables whose cell probabilities are in general pattern and whose dimensions may diverge. Specific examples are provided to demonstrate the asymptotic normality of test statistics when the classical regularity conditions for the normal approximation are violated. Simulation results support the theoretical findings and demonstrate that for testing independence and goodness-of-fit in large contingency tables, the degrees of freedom adjusted chi-squared or normal approximations are more robust than the traditional chi-squared approximation, especially when the table is sparse.

Keywords: Central limit theorem; Chi-squared approximation; Goodness-of-fit; Hoeffding's decomposition; Orthogonal projection; Poissonization

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1 Introduction

Statistical independence is a unique concept in probability theory that sets it apart from general measure theory. The test of independence is a fundamental problem in statistics, which underlies various applications and has been studied extensively by many researchers. Suppose a two-way contingency table $(X_{j,k})_{d_1\times d_2}$ is observed and modeled to follow a multinomial distribution with sample size n and cell probabilities $\theta_{j,k}$ in d_1 rows and d_2 columns. Pearson's chi-squared and the likelihood ratio tests are the best known statistical methods for testing the independence of the row and column effects in such a two-way contingency table. In the classic settings where d_1 and d_2 are fixed and the expected cell counts $n\theta_{j,k}$ diverge, Pearson's chi-squared statistic converges weakly to the chi-squared distribution χ^2_ν with degrees of freedom $\nu = (d_1 - 1)(d_2 - 1)$ under the null hypothesis of independence (Fisher, 1924). However, these conditions are often violated in large-scale applications dealing with high-dimensional and sparse data. New asymptotic theory which can handle the situation when $d_1 \times d_2$ diverges and $n\theta_{j,k} \to 0$ is anticipated. More importantly, by allowing the dimensions of the contingency table to diverge to infinity, one naturally extends the asymptotic theory for test of independence of categorical random variables to the test of independence of continuous random vectors via discretization procedures.

The measurement of association and test of independence among random vectors, dating back at least to Spearman (1904), are still very active areas of ongoing research. Miller and Siegmund (1982) studied the maximally selected chi-squared statistic by partitioning the range of data. Székely et al. (2007) proposed a statistic to test the independence of a pair of random vectors based on the empirical distance covariance, a weighted distance between the joint empirical characteristic function and the product of marginal empirical characteristic functions. By embedding the data into a reproducing kernel Hilbert space via a feature map, Gretton et al. (2007) proposed kernel based Hilbert-Schmidt independence criterion for testing pairwise independence. Pfister et al. (2018) generalized their method to test of the mutual independence of more than two random variables. Generalized measures of correlation were studied by Zheng et al. (2012) to deal with asymmetric and non-linear correlations between random variables. Through a binary expansion methodology, Zhang (2019) studied binning based multiple testing method for two-way contingency tables to detect local dependence. A ranking based correlation coefficient was proposed by Chatterjee (2021) and further discussed in Auddy et al. (2021), Bickel (2022) and Lin and Han (2023).

We provide necessary and sufficient conditions for the χ^2_ν approximation of the null distribution of Pearson's chi-squared statistic for the test of independence in two-way contingency tables $(X_{j,k})_{d_1\times d_2}$ with possibly diverging degrees of freedom $\nu = (d_1 - 1)(d_2 - 1)$. In the high-dimensional regime where $\nu \to \infty$, we prove that the χ^2_ν approximation is valid if and only if the expected number of cells with $X_{j,k} \geq 2$ diverges in any deterministic subset of cells occupying a constant fraction of the total $d_1 d_2$ cells, provided that the fractions of empty rows and columns in the table converge to zero and the variance of Pearson's chi-squared statistic is within an infinitesimal fraction of 2ν.

Traditional guidelines for the adequacy of χ^2_ν approximation suggest that the minimum expected frequency should not be too small, e.g., $\min_{j,k} n\theta_{j,k} \geq 5$, or $\min_{j,k} n\theta_{j,k} \geq 1 + o(1)$ and $n\theta_{j,k} < 5$ for no more than 20% of all cells (Cochran, 1952, 1954; Agresti, 2012), which entail a stringent sample size requirement $n \gtrsim d_1 d_2$. It was also noted in the literature that such guidelines were usually excessively restrictive (van der Waerden, 1971; Stahel, 2002). In contrast with the traditional guidelines, our seemingly counter-intuitive result implies that, when the χ^2_ν approximation holds in large two-way contingency tables, one may see an overwhelming fraction of empty cells with $X_{j,k} = 0$ (for example $n \ll d_1 d_2$), but one must expect to see a diverging number of cells with $X_{j,k} \geq 2$.

We propose the use of a more robust cross-sample version of Pearson's chi-squared statistic for the test of independence, for which the side condition on the variance is removed from the premises of above theory and the side condition on the empty rows and columns is also necessary. We propose a degrees-of-freedom adjusted chi-squared approximation to significantly gain size accuracy by matching Pearson's chi-squared or

the cross-sample chi-squared statistics in both the conditional mean and variance given the marginal totals. By matching both the mean and the variance, this approximation scheme allows a continuous transition from the standardized $(\chi^2_\nu - \nu)/(2\nu)^{1/2}$ approximation to the normal $\mathcal{N}(0,1)$ approximation throughout the low- and high-dimensional regimes as $\nu \to \infty$.

The above theory on the test of independence in two-way contingency tables is based on new results developed in this paper for Pearson's χ^2 goodness-of-fit test. Pearson's χ^2 and the likelihood ratio tests are widely used statistical methods for the test of goodness-of-fit in one-way contingency tables. Let $(\theta_1, \ldots, \theta_d)$ be a probability vector with $\theta_j > 0$ and $\sum_{j=1}^d \theta_j = 1$, and $(X_1, \ldots, X_d)$ be a random vector with the multinomial$(n, \theta_1, \ldots, \theta_d)$ distribution. With fixed dimension d and cell probabilities θ_j, Pearson's χ^2 statistic converges weakly to the chi-squared distribution with degrees of freedom $d-1$ as the sample size n diverges to infinity (Pearson, 1900). A similar conclusion, generally known as Wilks' phenomenon, holds for the log likelihood ratio. With a diverging d or varying θ_j's, the validity of the χ^2_{d-1} approximation is under question. For instance, Cochran (1952, p. 334) recommended the use of normal approximation instead when $d > 60$ and $\max_{1\le j\le d} n\theta_j < 5$. Various sets of sufficient conditions for the normal approximation of Pearson's χ^2 and the log likelihood ratio have appeared in the literature (Holst, 1972; Morris, 1975). Still, despite a long history of wide practical applications and theoretical investigations, the chi-squared and normal approximations of these statistics in large contingency tables are not completely understood.

Recent work of Xu et al. (2019) provided a distributional approximation of Pearson's χ^2 statistic in a dichotomous form: The asymptotic distribution can be either a chi-squared or a normal distribution, depending on regularity conditions, so that the asymptotic behavior of Pearson's χ^2 statistic may have a phase transition phenomenon in the high-dimensional regime. In a related multivariate Gaussian case, He et al. (2021) derived phase transition boundaries for the Wilks' phenomenon for likelihood ratio tests for a number of null hypotheses. Even in these specific settings, the approximation theory for the log likelihood ratio is still an uncharted land beyond the border of Wilks' phenomenon. In any case, necessary and sufficient conditions for the chi-squared and normal approximations of Pearson's χ^2 and the log likelihood ratio have still eluded existing studies in the literature.

We provide a complete characterization of the chi-squared and normal approximations of these statistics. For d and θ_j's depending on n in general pattern, we derive necessary and sufficient conditions for the χ^2_{d-1} and normal approximations of Pearson's χ^2 statistic as well as necessary and sufficient conditions for the normal approximation of the log likelihood ratio and Hellinger statistic. For example, for testing the equiprobable null hypothesis using either Pearson's χ^2 statistic or the log likelihood ratio, Wilks' phenomenon holds if and only if $n^2/d \to \infty$, and the normal approximation holds if and only if $d \wedge (n^2/d) \to \infty$. Consequently, the sufficient condition $\sqrt{d} = o(n)$, obtained in Xu et al. (2019) and suggested by Koehler and Larntz (1980) in a simulation study, is also necessary for the χ^2_{d-1} approximation of Pearson's χ^2 statistic. Moreover, as in the test of independence, a degrees-of-freedom adjusted chi-squared approximation applies continuously throughout the low- and high-dimensional regimes. Throughout the paper, $[d] = \{1, \ldots, d\}$ and $a_{1:d} = (a_1, \ldots, a_d)^\top$ for any positive integer d and reals a_j.

2 Chi-squared statistics

2.1 Test of independence

We consider the test of independence of the row and column effects in a large two-way contingency table. Let $X = (X_{j,k})_{d_1 \times d_2}$ be the data matrix with $X_{j,k}$ being the count for the cell in the j-th row and k-th column. The row, column and grand totals of the matrix X are given by

$$R_j = \sum_{k=1}^{d_2} X_{j,k}, \quad C_k = \sum_{j=1}^{d_1} X_{j,k}, \quad n = \sum_{j=1}^{d_1} R_j = \sum_{k=1}^{d_2} C_k. \tag{1}$$

Formally, we state the null hypothesis in the test of independence as

$$H_0 : \ X = (X_{j,k}, j \in [d_1], k \in [d_2]) \sim \text{multinomial}(n, p_j q_k, j \in [d_1], k \in [d_2]), \tag{2}$$

which means the independence of row and column effects, where $p_1, \ldots, p_{d_1}$ and $q_1, \ldots, q_{d_2}$ are treated as completely unknown parameters satisfying $p_j > 0$, $\sum_{j=1}^{d_1} p_j = 1$, $q_k > 0$, and $\sum_{k=1}^{d_2} q_k = 1$. Throughout the section, n and (d_1, d_2) are treated as deterministic and $d_1 \vee d_2$ are allowed to diverge as $n \to \infty$.

The classic chi-squared test of independence rejects the null hypothesis H_0 in (2) when the following statistic exceeds a certain threshold:

$$\chi^2_{\text{indep}} = \sum_{R_j>0} \sum_{C_k>0} \frac{(X_{j,k} - R_j C_k/n)^2}{R_j C_k/n} = \sum_{R_j>0} \sum_{C_k>0} \frac{X^2_{j,k}}{R_j C_k/n} - n. \tag{3}$$

The typical practice is to reject H_0 when $\chi^2_{\text{indep}} > t_\alpha$, where t_α is the $(1-\alpha)$-th quantile of the statistic determined by the hypergeometric distribution for $d_1 = d_2 = 2$, known as Fisher's exact test, and approximately determined by the χ^2_ν distribution with degrees of freedom $\nu = (d_1 - 1)(d_2 - 1)$ for more general d_1 and d_2. However, in the case where ν is allowed to diverge as $n \to \infty$, the validity of the χ^2_ν approximation is unclear in the existing literature and nontrivial as our analysis attests.

To best present our results for the χ^2_ν approximation of the above test statistic, we shall first consider the following cross-sample version of it,

$$\chi^2_{\text{CS-indep}} = \sum_{R_j>0} \sum_{C_k>0} \frac{X_{j,k}(X_{j,k} - 1)}{R_j C_k/n} - \frac{(n - d_1')(n - d_2')}{n - 1} \tag{4}$$

with $d_1' = \#\{j : R_j > 0\}$ and $d_2' = \#\{k : C_k > 0\}$. This test statistic for independence is motivated by the works of Bai and Saranadasa (1996) on Hotelling's T^2 test and Xu et al. (2019) on Pearson's goodness-of-fit test as we will discuss in Section 2.3 below. Write $X_{j,k} = \sum_{i=1}^n B_{i,j,k}$ with Bernoulli variables $B_{i,j,k}$. We call (4) the cross-sample version of the classic chi-squared statistic (3) because $X_{j,k}(X_{j,k} - 1) = \sum_{1 \le i \ne i' \le n} B_{i,j,k} B_{i'j,k}$. In addition, (4) is conditionally centered under H_0:

$$\mathbb{E}\Big[\chi^2_{\text{CS-indep}} \Big| R_{1:d_1}, C_{1:d_2}\Big] = 0.$$

The following theorem provides necessary and sufficient conditions for the χ^2_ν approximation of the sampling distribution of this cross-sample test statistic in two-way contingency tables.

Theorem 2.1. *Suppose (2) holds with $d_1 \wedge d_2 \ge 2$. Then, the chi-squared approximation*

$$\sup_{-\infty \le t \le \infty} \Big| \mathbb{P}\Big\{\chi^2_{\text{CS-indep}} \le t\Big\} - \mathbb{P}\Big\{\chi^2_\nu - \nu \le t\Big\}\Big| \to 0 \tag{5}$$

holds with $\nu = (d_1 - 1)(d_2 - 1)$ for the cross-sample chi-squared statistic in (4) if and only if

$$\frac{1}{d_1 d_2} \sum_{j=1}^{d_1} \sum_{k=1}^{d_2} \left(\frac{1}{1 + np_j} + \frac{1}{1 + nq_k} + \frac{1}{1 + \sqrt{d_1 d_2} np_j q_k} \right) \to 0. \tag{6}$$

Moreover, for $\nu \to \infty$, (6) holds if and only if the following two conditions hold:

$$\mathbb{E}\left[\frac{\#\{j \in [d_1] : R_j = 0\}}{d_1}\right] + \mathbb{E}\left[\frac{\#\{k \in [d_2] : C_k = 0\}}{d_2}\right] \to 0, \tag{7}$$

and for any deterministic index sets $S_n \subset [d_1] \times [d_2]$,

$$d_1 d_2 / |S_n| = O(1) \quad \Rightarrow \quad \mathbb{E}\big[\#\big\{(j,k) \in S_n : X_{j,k} \ge 2\big\}\big] \to \infty. \tag{8}$$

The parameters in Theorem 2.1 are all allowed to depend on the sample size n, $p_j = p_{j,n}$, $j \in [d_1]$, and $q_k = q_{k,n}$, $k \in [d_2]$, with $d_1 = d_{1,n}$ and $d_2 = d_{2,n}$. Thus, the theorem is uniform in $\{d_1, d_2, p_{1:d_1}, q_{1:d_2}\}$. For $\nu = (d_1 - 1)(d_2 - 1) = O(1)$, (6) is equivalent to the classic condition $\min_{j\in[d_1],k\in[d_2]} \mathbb{E}[X_{j,k}] \to \infty$. For $\nu \to \infty$, (7) and (8) give the statistical interpretation of the necessary and sufficient condition: The fractions of empty rows and columns converge to zero, and the expected number of cells with $X_{j,k} \geq 2$ diverges in any deterministic fractional subsets of cells.

Based on the analysis leading to the above theorem we develop adjusted degrees of freedom to improve the accuracy of the chi-squared approximation. This adjusted degrees of freedom is one half of the conditional variance of $\chi^2_{\text{CS-indep}}$ given the marginal totals $R_{1:d_1}$ and $C_{1:d_2}$ or the approximation of the conditional variance in the following proposition.

Proposition 2.2. *Let $\chi^2_{\text{CS-indep}}$ be as in (4). Then, under (2), $\mathbb{E}[\chi^2_{\text{CS-indep}}|R_{1:d_1}, C_{1:d_2}] = 0$ and the half conditional variance $\overline{v}_{\text{CS}} = \text{Var}(\chi^2_{\text{CS-indep}}|R_{1:d_1}, C_{1:d_2})/2$ can be expressed as*

$$\overline{v}_{\text{CS}} = \nu^* + \epsilon_n, \tag{9}$$

where

$$\nu^* = \frac{(d_1'' - 1)(d_2'' - 1) + (d_1' d_2' - 1)/n}{(1 - 1/n)(1 - 2/n)(1 - 3/n)} \tag{10}$$

with the d_1' and d_2' in (4), $d_1'' = \sum_{R_j>0}(1 - 1/R_j)$ and $d_2'' = \sum_{C_j>0}(1 - 1/C_j)$, and

$$\epsilon_n = \frac{(3\Delta_1\Delta_2 - n\Delta_1 - n\Delta_2)d_1' d_2' - n(d_1' + d_2' - 1)^2 - (d_1' - 1)^2(d_2' - 1)^2/(1 - 1/n)}{(n - 1)(n - 2)(n - 3)},$$

with $\Delta_m = n(1 - d_m''/d_m') - d_m' \geq 0$, $m = 1, 2$. Moreover,

$$-\frac{2 - d_1''/d_1' - d_2''/d_2' - 1/n}{(n - 1)(1 - 2/n)(1 - 3/n)} \leq \frac{\epsilon_n}{d_1' d_2'} \leq \frac{(1 - d_1''/d_1')(1 - d_2''/d_2')}{(n - 1)(1 - 2/n)(1 - 3/n)}$$

for $n \geq d_1 \wedge d_2$, and $\epsilon_n \leq 0$ when $\Delta_1 + \Delta_2 \leq (4/3)n$.

Proposition 2.2 provides a formula for the conditional variance and a second order approximation of it with an explicit error bound. This second order approximation ν^* gives highly accurate results for the type I error rates in our simulation experiments as reported in Section 4.1. By definition,

$$\Delta_1 = \sum_{R_j>0} \left(\frac{n}{R_j d_1'} + \frac{R_j d_1'}{n} - 2 \right), \quad \Delta_2 = \sum_{C_k>0} \left(\frac{n}{C_k d_2'} + \frac{C_k d_2'}{n} - 2 \right). \tag{11}$$

We propose to use ν^* in (10) as adjusted degrees of freedom to the conventional degrees of freedom $\nu = (d_1 - 1)(d_2 - 1)$. The following theorem provides necessary and sufficient conditions for the validity of the degrees-of-freedom adjustment with ν replaced by ν^* or $\overline{v}_{\text{CS}}$. Let $\chi^2_{\nu,1-\alpha}$ be the $(1 - \alpha)$-th quantile of the χ^2_ν distribution, with $\chi^2_\nu \sim \text{gamma}(\nu/2, 1/2)$ for all real $\nu > 0$.

Theorem 2.3. *Let ν^* and ϵ_n be as in Proposition 2.2. Suppose (2) and (7) hold. Then,*

$$\sup_{0\leq\alpha\leq1} \left| \mathbb{P}\left\{ \chi^2_{\text{CS-indep}} \geq \chi^2_{\widehat{\nu},1-\alpha} - \widehat{\nu} \right\} - \alpha \right| \to 0 \tag{12}$$

with either $\widehat{\nu} = \nu^*$ or $\widehat{\nu} = \overline{v}_{\text{CS}}$ if and only if (5) holds with $\nu = (d_1 - 1)(d_2 - 1)$, if and only if

$$\frac{1}{d_1 d_2} \sum_{j=1}^{d_1} \sum_{k=1}^{d_2} \frac{1}{1 + \sqrt{d_1 d_2} n p_j q_k} \to 0. \tag{13}$$

Moreover, for $\nu \to \infty$, (13) is equivalent to (8).

With the degrees-of-freedom adjustment $\widehat{\nu}$, the $\chi^2_{\widehat{\nu}}$ approximation (12) matches the cross-sample chi-squared statistic in both the mean and the conditional variance, which clearly provides more accurate results in our simulation experiments as reported in Section 4.1. In this sense, the degrees-of-freedom adjusted version (12) is more robust than (5).

Because (7) is necessary for (5), the degrees-of-freedom adjusted chi-squared approximation always holds as in (12) when the classic chi-squared approximation holds as in (5). However, when (7) does not hold, the upper limit of $\mathbb{E}[\nu^*/\nu]$ is strictly less than 1, so that (5) must fail. Condition (7) holds if and only if

$$\frac{1}{d_1}\sum_{j=1}^{d_1}\frac{1}{1+np_j}+\frac{1}{d_2}\sum_{k=1}^{d_2}\frac{1}{1+nq_k}=o(1), \tag{14}$$

which implies $n \gg d_1 \vee d_2$ by Jensen's inequality. However, Theorem 2.1 allows $n \ll d_1 d_2$, so that zero may occupy a majority of the cells in the contingency table. Compared with (13), condition (8) is expressed in terms of the counts of the cells with $X_{j,k} \geq 2$. Thus, when the chi-squared approximation holds in a large two-way contingency table, one must expect to see a diverging number of cells with $X_{j,k} \geq 2$, but one may still see an overwhelming fraction of cells with $X_{j,k} = 0$.

Conditionally on marginal totals $R_{1:d_1} = r_{1:d_1}$ and $C_{1:d_2} = c_{1:d_2}$, the data matrix X has the hypergeometric distribution under the null hypothesis (2) with joint probability mass function

$$\overline{p}(x)=\frac{(r_1!\cdots r_{d_1}!)(c_1!\cdots c_{d_2}!)}{n!\prod_{j=1}^{d_1}\prod_{k=1}^{d_2}x_{j,k}!} \tag{15}$$

for the counting matrix $x = (x_{j,k})_{d_1\times d_2}$ with $r_j = \sum_{k=1}^{d_2} x_{j,k}$ and $c_k = \sum_{j=1}^{d_1} x_{j,k}$.

We are now ready to study the classic test statistic for independence in (3). This is done by comparing it with its cross-sample version in (4) through the following proposition.

Proposition 2.4. *Let χ^2_{indep} be Pearson's chi-squared statistic in (3) for the test of independence and $\chi^2_{\text{CS-indep}}$ be its cross-sample version in (4). Suppose (2) holds. Then,*

$$\mathbb{E}\left[\chi^2_{\text{indep}}\middle|R_{1:d_1}, C_{1:d_2}\right]=\frac{(d_1'-1)(d_2'-1)}{1-1/n} \tag{16}$$

with the d_1' and d_2' in (4), and the half conditional variance $\overline{\nu} = \text{Var}(\chi^2_{\text{indep}}|R_{1:d_1}, C_{1:d_2})/2$ satisfies

$$\overline{\nu}=\overline{\nu}_{\text{CS}}+\frac{1}{2}\left(1+\frac{4}{n-2}\right)\overline{V}_{\text{diff}}, \tag{17}$$

where $\overline{V}_{\text{diff}} = \text{Var}(\chi^2_{\text{indep}} - \chi^2_{\text{CS-indep}}|R_{1:d_1}, C_{1:d_2}) = \Delta_1\Delta_2 d_1' d_2'/(n-1)$ with Δ_m $(m = 1, 2)$ in (11).

Proposition 2.4 gives an approximate orthogonal decomposition of Pearson's chi-squared statistic for the test of independence, as the difference $\chi^2_{\text{indep}} - \chi^2_{\text{CS-indep}}$ is nearly orthogonal to $\chi^2_{\text{CS-indep}}$. If (7) holds, $\text{Var}(\chi^2_{\text{CS-indep}}) = (1 + o(1))\nu$ by Proposition 2.2, so that $\overline{V}_{\text{diff}}$ is a small fraction of ν when

$$\text{Var}(\chi^2_{\text{indep}}) \leq (2 + o(1))\nu. \tag{18}$$

This leads to the following theorem.

Theorem 2.5. *Suppose (2), (7) and (18) hold with $\nu = (d_1 - 1)(d_2 - 1)$. Then,*

$$\sup_{-\infty\leq t\leq\infty}\left|\mathbb{P}\left\{\chi^2_{\text{indep}}-\frac{(d_1'-1)(d_2'-1)}{1-1/n}\leq t\right\}-\mathbb{P}\left\{\chi^2_\nu-\nu\leq t\right\}\right|\to 0 \tag{19}$$

if and only if (13) *holds, if and only if* (8) *holds, if and only if*

$$\sup_{0\le\alpha\le 1}\left|\mathbb{P}\left\{\chi^2_{\text{indep}} - \frac{(d_1'-1)(d_2'-1)}{1-1/n} \ge \chi^2_{\widehat{\nu},1-\alpha} - \widehat{\nu}\right\} - \alpha\right| \to 0 \tag{20}$$

with $\widehat{\nu} = (d_1'-1)(d_2'-1)$ *or* $\widehat{\nu} = \overline{\nu}$, *where* $\overline{\nu}$ *is the half conditional variance in* (17).

It follows from Theorems 2.3 and 2.5 that (19), (20), (5) and (12) are all equivalent under (2), (7) and (18). However, the side condition (18), required only for the classic Pearson's chi-squared statistic, is noticeably more restrictive because approximately $\overline{V}_{\text{diff}}$ is expected to be

$$V_{\text{diff}} = \frac{d_1 d_2}{n}\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\left(\frac{1}{d_1 p_j} + d_1 p_j - 2\right)\left(\frac{1}{d_2 q_k} + d_2 q_k - 2\right)$$

under conditions (2) and (7). For example, when $p_j = (1\pm\epsilon)/d_1$ and $q_k = (1\pm\epsilon)/d_2$ with even d_1 and d_2 and $\epsilon = 1/\sqrt{2}$, $V_{\text{diff}} = (d_1 d_2)^2/n$, so that the requirement $V_{\text{diff}} = o(\nu)$ entails that $n \gg d_1 d_2$. From this point of view, the chi-squared approximation in (5) and (12) for the cross-sample statistic are more robust than their counterpart in (19) and (20) for the classic Pearson's chi-squared statistic.

Based on the results of this subsection, the approximation (12) is expected to be most accurate amongst the four methods. This is supported by our simulation results reported in Section 4.1. The proofs of Theorems 2.1, 2.3 and 2.5 are based on our study below of the one-way contingency table.

2.2 Power analysis for the cross sample version of chi-squared test of independence

We consider a probability $\mathbb{P}$ under which $X = (X_{j,k})_{d_1\times d_2}$ ~multinomial$(n, (\theta_{j,k})_{d_1\times d_2})$, and the null cell probabilities are $\kappa_{j,k} = p_j q_k$, $j \in [d_1]$, $k \in [d_2]$, where $p_j = \sum_{k=1}^{d_2}\theta_{j,k}$ and $q_k = \sum_{j=1}^{d_1}\theta_{j,k}$ are respectively the row and column marginal probabilities. We shall focus on the more stable cross-sample version of the test.

If the marginal probabilities p_j, q_k were known, the cross-sample chi-squared test statistic boils down to

$$\chi^2_{\text{CS}} = \sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\frac{X^{(2)}_{j,k}}{np_j q_k} - (n-1). \tag{21}$$

Under $\mathbb{P}$, this χ^2_{CS} has mean

$$\begin{aligned}\mu_{\text{CS}} \;=\; & (n-1)\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}(\rho_{j,k}-1)^2 p_j q_k - \sum_{j=1}^{d_1}(\overline{\rho}_{j,*}-1)\big(\mathbb{P}\{R_j>0\} - p_j\big)\\ & - \sum_{k=1}^{d_w}(\overline{\rho}_{*,k}-1)\big(\mathbb{P}\{C_k>0\} - q_k\big) + \sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\rho^2_{j,k}(\theta_{j,k} - p_j q_k)\end{aligned} \tag{22}$$

with $\rho_{j,k} = \theta_{j,k}/(p_j q_k)$, $\overline{\rho}_{j,*} = \sum_{k=1}^{d_2}\rho_{j,k}\theta_{j,k}/p_j$ and $\overline{\rho}_{*,k} = \sum_{j=1}^{d_1}\rho_{j,k}\theta_{j,k}/q_k$, and variance

$$V_{\text{CS}} = \sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\frac{\sigma^2_{\text{CS},j,k}}{n} + \big(1-1/n\big)\big(V^*_{\text{CS}} - V^*_{\text{row}} - V^*_{\text{col}} + V^*_{rc}\big) \tag{23}$$

with $\sigma_{\text{CS},j,k} = 2(n-1)(\rho_{j,k}-\overline{\rho})\theta^{1/2}_{j,k}$, $\overline{\rho} = \sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\rho_{j,k}\theta_{j,k}$,

$$V^*_{\text{CS}} = 2\big(\overline{\rho}^2 + \textstyle\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\rho^2_{j,k}(1-2\theta_{j,k})\big),$$

$$V^*_{\text{row}} = 4\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\rho_{j,k}^2\theta_{j,k}(1/p_j - 1) - 2\sum_{j=1}^{d_1}\rho_{j,*}^2 + 2\overline{\rho}^2,$$
$$V^*_{\text{col}} = 4\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\rho_{j,k}^2\theta_{j,k}(1/q_k - 1) - 2\sum_{k=1}^{d_2}\rho_{*,k}^2 + 2\overline{\rho}^2,$$

and $V^*_{rc} = V'_{rc} + 4V''_{rc} - 2V'''_{rc} + V''''_{rc}$ with

$$V'_{rc} = \sum_{j=1}^{d_1}\sum_{k=1}^{d_2}(\rho_{j,k}^2 - \overline{\rho}_{*,k})(\rho_{j,k}^2 - \overline{\rho}_{j,*})p_jq_k,$$
$$V''_{rc} = \sum_{j=1}^{d_1}\sum_{k=1}^{d_2}(\rho_{j,k} - \overline{\rho}_{*,k})(\rho_{j,k} - \overline{\rho}_{j,*})(\rho_{j,k} - 1)\theta_{j,k},$$
$$V'''_{rc} = \sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\overline{\rho}_{j,*}\overline{\rho}_{*,k}(\rho_{j,k} - 1)^2p_jq_k,$$
$$V''''_{rc} = \sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\sum_{j'=1}^{d_1}\sum_{k'=1}^{d_2}\rho_{j,k}^2\rho_{j',k'}^2(\theta_{j,k'} - p_jq_{k'})(\theta_{j',k} - p_{j'}q_k).$$

If the null hypothesis is true, or equivalently $\theta_{j,k} = p_jq_k$ ($\rho_{j,k} = 1$), for all j, k, we have $\mu_{\text{CS}} = 0$ and $V_{\text{CS}} = 2(1 - 1/n)(d_1d_2 - 1)$. When $\rho_{j,k} \approx 1$, the mean and variance of χ^2_{CS} approximately match those of a non-central chi-squared distribution with $\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\sigma^2_{\text{CS},j,k}/n \approx 4\mu_{\text{CS}}$ as the contribution of the non-centrality to the variance. See section 2.4 for a power analysis of χ^2_{CS}.

In the case of most interest where the marginal probabilities are unknown, we approximate p_jq_k in (21) by R_jC_k/n^2 in the test statistic $\chi^2_{\text{CS-indep}}$ in (4). As a result, the test statistic is not a polynomial of the cell counts $X_{j,k}$, and the mean and variance of $\chi^2_{\text{CS-indep}}$ do not have simple expressions when $\theta_{j,k} \neq p_jq_k$. Thus, under (possibly alternative) probability $\mathbb{P}$, we approximate the mean of $\chi^2_{\text{CS-indep}}$ by

$$\mu_{\text{CS-indep}} = \sum_{j=1}^{d_1}\sum_{k=1}^{d_2}(\rho_{j,k} - 1)^2\Big((n+1)p_jq_k - q_k\mathbb{P}\{R_j > 0\} - p_j\mathbb{P}\{C_k > 0\}\Big), \tag{24}$$

and approximate the variance of $\chi^2_{\text{CS-indep}}$ by

$$\begin{aligned} V_{\text{CS-indep}} \quad = \quad & \sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\frac{\sigma^2_{j,k}}{n} + 2\Big(1 - \frac{1}{n}\Big)\Big(\overline{\rho}^2 + \sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\rho_{j,k}^2(1 - 2\theta_{j,k})\Big) \\ & -2\Big(1 - \frac{1}{n}\Big)\Big(2\overline{\rho}^2 + \sum_{j=1}^{d_1}\overline{\rho}^2_{j,*}(1 - 2p_j) + \sum_{k=1}^{d_2}\overline{\rho}^2_{*,k}(1 - 2q_k)\Big) \\ & +4\Big(1 - \frac{1}{n}\Big)\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\overline{\rho}_{j,*}\overline{\rho}_{*,k}(\rho_{j,k} - 1)^2p_jq_k, \end{aligned} \tag{25}$$

where $\overline{\rho}_{j,*} = \sum_{k=1}^{d_2}\rho_{j,k}^2q_k$, $\overline{\rho}_{*,k} = \sum_{j=1}^{d_1}\rho_{j,k}^2p_j$, and $\sigma_{j,k} = (n-1)(2\rho_{j,k} - \overline{\rho}_{j,*} - \overline{\rho}_{*,k})\theta_{j,k}^{1/2}$. In fact, $\mu_{\text{CS-indep}}$ and $V_{\text{CS-indep}}$ are the mean and variance of a quadratic function of $X_{j,k}$ as an approximation of the test statistic $\chi^2_{\text{CS-indep}}$. Both the approximate mean (24) and variance (25) exhibit row and column effects due to the use of estimated marginal probabilities in (21), leading to smaller values compared with (22) and (23) respectively. Under the null hypothesis of $\rho_{j,k} = 1, \forall\, j, k$, the above expressions reduce to $\mu_{\text{CS-indep}} = 0$ and $V_{\text{CS-indep}} = 2(1 - 1/n)(d_1 - 1)(d_2 - 1)$.

Consider the assumption below

$$\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\rho_{j,k}^2\theta_{j,k} + \sum_{k=1}^{d_2}\overline{\rho}^2_{*,k} + \max_{1\le j\le d_1}\overline{\rho}^2_{j,*} \ll V_{\text{CS-indep}}, \quad \sum_{j=1}^{d_1}\overline{\rho}^2_{j,*} \lesssim V_{\text{CS-indep}}. \tag{26}$$

For general $\rho_{j,k}$, (26) implies

$$V_{\text{CS-indep}} \asymp V = \sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\left(\frac{\sigma^2_{j,k}}{n} + 2\rho^2_{j,k}\right). \tag{27}$$

Under the null hypothesis $\rho_{j,k} = 1, \forall j, k$, (26) is equivalent to $2 \le d_2$ and $d_1 \to \infty$.

In addition, similar to (14), we impose the following side conditions:

$$\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\left(\frac{\rho_{j,k}^2}{1+n(p_j \wedge q_k)} + \left(\rho_{j,k}^2 - \overline{\rho}_{j,*} - \overline{\rho}_{*,k} + \overline{\rho}\right)^2 p_j q_k\right) \ll V_{\text{cs-indep}},$$
$$\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\left((\rho_{j,k} - \overline{\rho}_{j,*})^2 \frac{\theta_{j,k}}{p_j} + (\rho_{j,k} - \overline{\rho}_{*,k})^2 \frac{\theta_{j,k}}{q_k}\right) \ll V_{\text{cs-indep}}, \quad \overline{\rho} = O(1). \tag{28}$$

Under the null hypothesis $\rho_{j,k} = 1, \forall j, k$, (28) is equivalent to (14).

Theorem 2.6. *Let $\chi^2_{\text{cs-indep}}$ be as in (4) with data $X = (X_{j,k})_{d_1 \times d_2}$. Suppose $X \sim \text{multinomial}(n, (\theta_{j,k})_{d_1 \times d_2})$ under a (possibly alternative) probability $\mathbb{P}$. Suppose (14), (26) and (28) hold and $d_1 d_2 \lesssim V$ for the quantity V in (27). Then,*

$$\sup_t \left|\mathbb{P}\left\{\chi^2_{\text{cs-indep}} - \mu_{\text{cs-indep}} \le t\right\} - \mathbb{P}\left\{\chi^2_{d'}(\tau') - (d' + \tau') \le t\right\}\right| \to 0 \tag{29}$$

for some (d', τ') satisfyiing $2d' + 4\tau' \ge (1 + o(1))V$ if and only if $(2d' + 4\tau')/V \to 1$ and

$$\max_{j,k} \frac{\Delta_{j,k}^2 \lambda_{j,k} + \rho_{j,k}^2}{1 + \lambda_{j,k} \wedge (V/\rho_{j,k}^2)} + \sum_{j,k}\left(\frac{\Delta_{j,k}^4 \lambda_{j,k}}{\Delta_{j,k}^2 + V} + \frac{\rho_{j,k}^4}{\rho_{j,k}^2 + \lambda_{j,k}^2 V}\right) \ll V, \tag{30}$$

where $\Delta_{j,k} = 2\rho_{j,k} - \overline{\rho}_{j,*} - \overline{\rho}_{*,k}$, $\lambda_{j,k} = n\theta_{j,k}$, V is as in (27), and the maximum and summation are taken over $(j, k) \in [d_1] \times [d_2]$.

2.3 Pearson's goodness-of-fit test

In a one-way contingency table with data $X_{1:d}$, Pearson's χ^2 statistic is by definition

$$\chi^2 = \sum_{j=1}^{d} \frac{(X_j - n\theta_j)^2}{n\theta_j} = \sum_{j=1}^{d} \frac{X_j^2}{n\theta_j} - n \tag{31}$$

with $\theta_j > 0$ and $\sum_{j=1}^d \theta_j = 1$. The cross-sample version of χ^2, recently studied in Xu et al. (2019), is

$$\chi^2_{\text{cs}} = \sum_{j=1}^{d} \frac{X_j(X_j - 1)}{n\theta_j} - n + 1. \tag{32}$$

Define quantities σ_j, $j \in [d]$, $\sigma > 0$ and V by

$$\sigma_j = \theta_j^{1/2}(1/\theta_j - d), \quad \sigma^2 = \sum_{j=1}^{d} \sigma_j^2, \quad V = \sigma^2/n + 2(d-1)(1 - 1/n). \tag{33}$$

As we will explain in the proof, $\sigma^2/n = \text{Var}(\chi^2 - \chi^2_{\text{cs}})$, and $\chi^2 - \chi^2_{\text{cs}} = \sum_{j=1}^d X_j/(n\theta_j) - 1$, an average of i.i.d. random variables, is the linear part of Pearson's χ^2 statistic in its Hoeffding decomposition.

Theorem 2.7. *Let $(X_1, \ldots, X_d) \sim \text{multinomial}(n, \theta_1, \ldots, \theta_d)$, where $d = d_n \ge 2$ and $\theta_j = \theta_{j,n}$ for $j \in [d]$ are allowed to depend on n. For $\alpha > 0$, let χ^2_α denote a $\text{gamma}(\alpha/2, 1/2)$ random variable. Then, the mean*

and variance of Pearson's χ^2 in (31) are given by $\mathbb{E}\left[\chi^2\right] = d-1$ and $\mathrm{Var}\left(\chi^2\right) = V$. Moreover, with $d^* = V/2$ as adjusted degrees of freedom,

$$\sup_t \left| \mathbb{P}\left\{\chi^2 - (d-1) \le t\right\} - \mathbb{P}\left\{\chi^2_{d'} - d' \le t\right\} \right| \to 0 \tag{34}$$

for some $d' \ge (1+o(1))d^*$ if and only if the following two conditions hold: $d'/d^* \to 1$ and

$$\frac{1}{d^*} \max_{1\le j\le d} \frac{\sigma_j^2/n}{1+n\theta_j} + \frac{1}{d^*} \sum_{j=1}^{d} \left(\frac{\sigma_j^4/n^2}{\sigma_j^2/n + d^* n\theta_j} + \frac{1}{1+d^*(n\theta_j)^2} \right) \to 0. \tag{35}$$

The classic choice of d' is $d' = d-1$ to match the mean: $\mathbb{E}[\chi^2] = d-1$. This approximation is variance consistent if and only if $\sigma^2 \ll nd$. However, the condition $\sigma^2 \ll nd$ is nontrivial due to the singularity of σ^2 at small $n^{1/2}\theta_j$. Theorem 2.7 advocates the choice $d' = d^*$ to match the variance $\mathrm{Var}(\chi^2) = 2d^*$ after centering both Pearson's χ^2 and its approximation at the mean. This is a more natural approximation because $\chi^2 - (d-1)$ and $\chi^2_{d^*} - d^*$ match in both the mean and variance. For this reason, we call d^* adjusted degrees of freedom. Figure 1 in Section 3.3 demonstrates that $\mathrm{Var}(\chi^2)/2$ can be quite different from $\mathbb{E}[\chi^2]$. While (34) with $d' = d-1$ is commonly used, for general $d = d_n \to \infty$, the necessary and sufficient condition in Theorem 2.7 is new to the best of our knowledge.

As we will explain in the proof of Theorem 2.7, for fixed $d \ge 2$ or $2 \le d = d_n = O(1)$, condition (35) is equivalent to $\min_{1\le j\le d}(n\theta_j) \to \infty$ and implies $d^*/(d-1) \to 1$, so that its equivalence to (34) with $d' = d-1$ recovers the classical χ^2_{d-1} approximation theorem for Pearson's χ^2 statistic in (31).

Corollary 2.8. *For $\sigma^2 \ll nd$, the χ^2_{d-1} approximation*

$$\sup_t \left| \mathbb{P}\left\{\chi^2 \le t\right\} - \mathbb{P}\left\{\chi^2_{d-1} \le t\right\} \right| \to 0 \tag{36}$$

holds if and only if

$$\frac{1}{d} \sum_{j=1}^{d} \frac{1}{1+d^{1/2}n\theta_j} \to 0. \tag{37}$$

Corollary 2.8 can be seen as follows. Condition (36) implies $d^*/(d-1) = 1 - 1/n + \sigma^2/(n(d-1)) \to 1$ in view of Theorem 2.7. When $\sigma^2/(nd) \to 0$, (37) is equivalent to (35). Theorem 2.7 also provides, as a corollary, necessary and sufficient conditions for the normal approximation of Pearson's χ^2 statistic.

Corollary 2.9. *For $V' \ge (1+o(1))V$, the normal approximation*

$$\sup_t \left| \mathbb{P}\left\{\chi^2 - (d-1) \le t\right\} - \mathbb{P}\left\{\mathcal{N}(0,V') \le t\right\} \right| \to 0 \tag{38}$$

holds if and only if the following two conditions hold: $V'/V \to 1$ and

$$\frac{1}{d} + \frac{1}{d^*} \max_{1\le j\le d} \frac{\sigma_j^2/n}{1+n\theta_j} + \frac{1}{d^*} \sum_{j=1}^{d} \left(\frac{\sigma_j^4/n^2}{\sigma_j^2/n + d^* n\theta_j} + \frac{1}{1+d^*(n\theta_j)^2} \right) \to 0. \tag{39}$$

Corollary 2.9 can be seen as follows. For $d = O(1)$, the normal approximation does not hold as $\chi^2 \ge 0$. For $d \to \infty$, $(\chi^2_{d^*} - d^*)/V^{1/2} \xrightarrow{\mathrm{D}} \mathcal{N}(0,1)$ as $V = 2d^* \to \infty$. Thus, compared with (34), the normal approximation just needs one more condition $d \to \infty$.

The χ^2_{d-1} and normal approximations were recently studied by Xu et al. (2019), who proved (36) under the sufficient condition

$$\frac{\sigma^2}{nd} + \frac{1}{d}\sum_{j=1}^{d} \frac{1}{(d^{1/2} n\theta_j)^{\delta}} \to 0 \tag{40}$$

for some $0 < \delta \le 1$. For normal approximation, they also proved

$$\sup_t \left| \mathbb{P}\{\chi^2 - (d-1) \le t\} - \mathbb{P}\{\mathcal{N}(0, \sigma^2/n) \le t\} \right| \to 0 \tag{41}$$

under the sufficient condition

$$nd/\sigma^2 + \mathbb{E}\big[W_1^2 I\{W_1^2 \ge n^{1/2}\epsilon\}\big] \to 0 \tag{42}$$

with $W_1 = \sum_{j=1}^{d}(B_{1,j}/\theta_j - 1)/\sigma$ and $(B_{1,1}, \ldots, B_{1,d}) \sim$ multinomial$(1, \theta_1, \ldots, \theta_d)$. While the second component of (42) is also necessary for (38) in the regime $nd \ll \sigma^2$, it is not necessary in the regime $\sigma^2 \ll nd$. As discussed in Xu et al. (2019), their approximation theory for Pearson's χ^2 statistics presents a dichotomy of the χ^2_{d-1} and $N(d-1, \sigma^2/n)$ regimes, which creates some extra difficulties in real applications. Their remedy is to use (32) as a diagonal-removed version (Bai and Saranadasa, 1996) of (31). Our analysis provides necessary and sufficient conditions for the centered χ^2_{d-1} approximation of (32).

Theorem 2.10. *Let $\chi^2_{\rm cs}$ be as in (32) and $(X_1, \ldots, X_d) \sim$ multinomial$(n, \theta_1, \ldots, \theta_d)$ with $d = d_n \ge 2$ and $\theta_j = \theta_{j,n}$ as in Theorem 2.7. Then, $\mathbb{E}\big[\chi^2_{\rm cs}\big] = 0$ and $\mathrm{Var}\big(\chi^2_{\rm cs}\big) = 2(d-1)(1-1/n)$. Moreover,*

$$\sup_t \left| \mathbb{P}\Big\{\chi^2_{\rm cs} \le t\Big\} - \mathbb{P}\Big\{\chi^2_{d-1} - (d-1) \le t\Big\} \right| \to 0 \tag{43}$$

if and only if

$$\frac{1}{d}\sum_{j=1}^{d} \frac{1}{1 + d(n\theta_j)^2} \to 0, \tag{44}$$

if and only if for all $S_n \subseteq \{1, \ldots, d\}$,

$$\frac{d}{|S_n|} = O(1) \;\;\Rightarrow\;\; \sum_{j \in S_n} \mathbb{P}\big\{X_j \ge 2\big\} \to \infty. \tag{45}$$

2.4 Power analysis for chi-squared goodness-of-fit tests

Suppose $X \sim$multinomial$(n, \theta_{1:d})$ and we want to test its goodness-of-fit against a null distribution with probability vector $\kappa_{1:d}$. Let

$$\mu = \sum_{j=1}^{d} \rho_j - \overline{\rho} + n(\overline{\rho} - 1) \tag{46}$$

with $\rho_j = \theta_j/\kappa_j$ and $\overline{\rho} = \sum_{j=1}^{d} \rho_j \theta_j$, and

$$V = \sum_{j=1}^{d} \frac{\sigma_j^2}{n} + 2\left(1 - \frac{1}{n}\right)\left(\overline{\rho}^2 + \sum_{j=1}^{d} \rho_j^2(1 - 2\theta_j)\right) \tag{47}$$

with $\sigma_j = \{1/\kappa_j - \sum_{k=1}^{d} \rho_k + 2(n-1)(\rho_j - \overline{\rho})\}\theta_j^{1/2}$.

We call $\chi^2_\alpha(\tau)$ a non-central chi-square random variable if it has the density

$$\sum_{k=0}^{\infty} \frac{e^{-\tau/2}(\tau/2)^k}{k!} \frac{x^{\alpha/2+k-1}e^{-x/2}}{2^{\alpha/2+k}\text{gamma}(\alpha+2k)}, \quad x>0.$$

It is well understood that $\chi^2_\alpha(\tau)$ has mean $\alpha+\tau$ and variance $2\alpha+4\tau$.

Theorem 2.11. *Let χ^2 be Pearson's chi-squared statistic in (31) with data $X_{1:d}$ and null parameter $\kappa_{1:d}$. Suppose $X_{1:d} \sim$ multinomial$(n, \theta_{1:d})$ under a probability $\mathbb{P}$ with (possibly alternative) parameter $\theta_{1:d}$. Then, $\mathbb{E}[\chi^2] = \mu$ with the μ in (46) and $\text{Var}(\chi^2) = V$ with the V in (47). Moreover, when $\sum_{k=1}^d \rho_k^2\theta_k \ll V$,*

$$\sup_t \left|\mathbb{P}\{\chi^2 - \mu \le t\} - \mathbb{P}\{\chi^2_{d'}(\tau') - (d'+\tau') \le t\}\right| \to 0 \tag{48}$$

for some (d', τ') satisfying $2d'+4\tau' \ge (1+o(1))V$ if and only if $(2d'+4\tau')/V \to 1$ and

$$\max_{1\le j\le d} \frac{\sigma_j^2/n + \rho_j^2}{1+(n\theta_j)\wedge(V/\rho_j^2)} + \sum_{j=1}^d \left(\frac{\sigma_j^4/n^2}{\sigma_j^2/n + n\theta_j V} + \frac{\rho_j^4}{\rho_j^2 + (n\theta_j)^2 V}\right) \ll V. \tag{49}$$

where ρ_j and σ_j are as in (46) and (47) respectively.

When $\theta_{1:d} = \kappa_{1:d}$, $\rho_j = 1$ and $\sigma_j = (1/\kappa_j - d)\theta_j^{1/2}$, so that $\mu = d-1$ and $V = 2(1-1/n)(d-1) + \sigma^2/n$ are as in Theorem 2.7 and the side condition $\sum_{k=1}^d \rho_k^2\theta_k \ll V$ becomes $1 \ll V$. Thus, Theorem 2.11 yields Theorem 2.7 in the case of $d = d_n \to \infty$.

For the cross-sample version of χ^2, let

$$\mu_{\text{CS}} = (n-1)(\overline{\rho}-1) = (n-1)\sum_{j=1}^d (\rho_j - 1)^2\kappa_j. \tag{50}$$

with $\rho_j = \theta_j/\kappa_j$ and $\overline{\rho} = \sum_{j=1}^d \rho_j\theta_j$ as in (46), and

$$V_{\text{CS}} \;=\; \sum_{j=1}^d \frac{\sigma_{\text{CS},j}^2}{n} + 2\left(1-\frac{1}{n}\right)\left(\overline{\rho}^2 + \sum_{j=1}^d \rho_j^2(1-2\theta_j)\right). \tag{51}$$

with $\sigma_{\text{CS},j} = 2(n-1)(\rho_j - \overline{\rho})\theta_j^{1/2}$.

Theorem 2.12. *Let χ^2_{CS} be the cross-sample chi-squared statistic in (32) with data $X_{1:d}$ and null parameter $\kappa_{1:d}$. Suppose $X_{1:d} \sim$ multinomial$(n, \theta_{1:d})$ under a probability $\mathbb{P}$ with (possibly alternative) parameter $\theta_{1:d}$. Then, $\mathbb{E}[\chi^2_{\text{CS}}] = \mu_{\text{CS}}$ with the μ_{CS} in (50), and $\text{Var}(\chi^2_{\text{CS}}) = V_{\text{CS}}$ with the V_{CS} in (51). Moreover, under the condition $\sum_{k=1}^d \rho_k^2\theta_k \ll V_{\text{CS}}$,*

$$\sup_t \left|\mathbb{P}\{\chi^2_{\text{CS}} - \mu_{\text{CS}} \le t\} - \mathbb{P}\{\chi^2_{d'}(\tau') - (d'+\tau') \le t\}\right| \to 0 \tag{52}$$

for some (d', τ') satisfyiing $2d'+4\tau' \ge (1+o(1))V_{\text{CS}}$ if and only if $(2d'+4\tau')/V_{\text{CS}} \to 1$ and

$$\max_{1\le j\le d} \frac{n(\rho_j-\overline{\rho})^2\theta_j + \rho_j^2}{1+(n\theta_j)\wedge(V_{\text{CS}}/\rho_j^2)} + \sum_{j=1}^d \left(\frac{n(\rho_j-\overline{\rho})^4\theta_j}{(\rho_j-\overline{\rho})^2 + V_{\text{CS}}} + \frac{\rho_j^4}{\rho_j^2 + (n\theta_j)^2 V_{\text{CS}}}\right) \ll V_{\text{CS}}. \tag{53}$$

Again, Theorem 2.12 yields Theorem 2.10 in the case of $d = d_n \to \infty$.

We write Pearson's chi-squared statistic as a second order U-statistic as follows. Let $B_i = B_{i,1:d}$ be independent multinomial$(1, \theta_{1:d})$ vectors and $X = X_{1:d} = \sum_{i=1}^n B_i$. Let $M = (M_{j,k})_{d\times d}$ with $M_{j,k} = I\{j = k\}/\theta_j - 1$. Assume $\min_{1\le j\le d} \theta_j > 0$. We have

$$\chi^2 = \sum_{j=1}^d \frac{X_j^2}{n\theta_j} - n = \frac{X^\top M X}{n} = \sum_{i_1=1}^n \sum_{i_2=1}^n \frac{B_{i_1}^\top M B_{i_2}}{n}$$

with $\mathbb{E}[\chi^2] = d - 1$. Hoeffding's decomposition of χ^2 is

$$\chi^2 = \chi^2_{CS} + \sum_{j=1}^d \left(\frac{X_j}{n\theta_j} - 1\right) + (d-1)$$

where χ^2_{CS}, the cross-sample version of χ^2, is the degenerate second order component,

$$\chi^2_{CS} = \chi^2 - \sum_j \frac{X_j}{n\theta_j} + 1 = \sum_{1\le i_1\ne i_2\le n} \frac{B_{i_1}^\top M B_{i_2}}{n}. \tag{54}$$

We apply Theorem **??** to approximate the distribution of χ^2_{CS}. We note that

$$M\theta_{1:d} = 0, \quad \mathbb{E}[MB_1B_1^\top M] = M, \quad \mathrm{Var}(B_1^\top M B_2) = \mathbb{E}[B_1^\top M B_1] = d - 1, \tag{55}$$

due to $M\mathbb{E}[B_1B_1^\top]M = M(\mathrm{diag}(\theta_{1:d}) - \theta_{1:d}\theta_{1:d}^\top)M = M - (\theta_{1:d}, \dots, \theta_{1:d})^\top M = M$. Let

$$\psi(x, y) = \frac{x^T M y}{\sqrt{d-1}}. \tag{56}$$

We find that $\mathrm{Var}(\psi(B_1, B_2)) = 1$ by (55), and in (**??**) with $\varepsilon_i = B_i$ and $v_n = 1 - 1/n$

$$W = \sum_{1\le i_1\ne i_2\le n} \frac{\sqrt{v_n}\psi(B_{i_1}, B_{i_2})}{\sqrt{2n(n-1)}} = \frac{\chi^2_{CS}}{\sqrt{2(d-1)}}. \tag{57}$$

The v_n here gives the right normalization as $2(d-1)$ is the variance of the target χ^2_{d-1} distribution. It also gives $\mathrm{Var}(\chi^2_{CS})/(2(d-1)) = \mathrm{Var}(W) = v_n$. By (**??**) and (55),

$$\overline{\psi}(x, y) = \frac{\psi(x, y)}{\sqrt{d-1}} = \frac{x^T M y}{d-1}, \tag{58}$$

so that $(d-1)^{-1/2}$ is the common nonzero eigenvalue of the operator ψ. The quantity $V_{T,\alpha}$ is nearly minimized with the shape parameter $\alpha = (d-1)/2$ of χ^2_{d-1}, giving

$$\begin{aligned} & \mathbb{E}\left[\left(T - 1 - W/\sqrt{\alpha}\right)^2\right] \\ \le\ & \mathrm{Var}\left(\sqrt{2}(2/n)\overline{\psi}(B_1, B_2)\right) + \mathrm{Var}\left(\overline{\psi}(B_1, B_1)\right)/n + (v_n - 1)^2 \\ =\ & \frac{8}{n^2(d-1)} + \frac{1}{n}\sum_{j=1}^d \frac{1/\theta_j - d}{(d-1)^2} + \frac{1}{n^2}. \end{aligned}$$

As $\theta_j > 0$ for all $j \le d$, this yields $\mathbb{E}\left[\left|T - 1 - W/\sqrt{\alpha}\right|\right] \lesssim 1/\sqrt{n}$ in (**??**) for fixed $\theta_{1:d}$. For general $\theta_{1:d}$ with diverging dimension, bounding separately the rare cell contribution would provide sharper upper bound for $\mathbb{E}\left[\left|T - 1 - W/\sqrt{\alpha}\right|\right]$. Moreover, specialized to (54), the terms $\overline{R}_1$ and $\overline{R}_2$ in Theorem **??** can be sharpened using the following lemma. This is an outline of the analysis leading to the L_1 bound for the χ^2_ν approximation of χ^2_{CS} in Theorem 2.14.

Lemma 2.13. *Let $X_{1:d} \sim$ multinomial$(n, \theta_{1:d})$ with $d \geq 2$ and $\theta_j > 0\, \forall j \in [d]$. Then,*

$$\begin{aligned}
\mathbb{E}\big[|X_j - n\theta_j|^3\big] &\leq 2n^{3/2}\big(\theta_j(1-\theta_j)\big)^{3/2} + n\theta_j(1-\theta_j), \quad \forall j, \\
\mathbb{E}\big[|X_j - n\theta_j|^2 |X_k - n\theta_k|\big] &\leq n^{3/2}\left(\theta_j\sqrt{\theta_k} + \frac{\theta_j^2(\theta_k - \sqrt{\theta_k})}{1-\theta_k}\right) + \frac{n\theta_j^2\theta_k}{1-\theta_k}, \quad j \neq k, \\
\sum_{j=1}^{d}\sum_{k=1}^{d} \mathbb{E}\big[(X_j - n\theta_j)^2 |X_k - n\theta_k|\big]/\theta_j &\leq (d\sqrt{d-1}+1)n^{3/2} + dn.
\end{aligned}$$

Theorem 2.14. *Let $d \geq 2$, χ^2_{CS} be as in (54), $\nu = d-1$, and χ^2_ν be a random variable with the chi-squared distribution with ν degrees of freedom. Then,*

$$d_{L_1}\left(\frac{\chi^2_{CS}}{\sqrt{2\nu}}, \frac{\chi^2_\nu - \nu}{\sqrt{2\nu}}\right) \leq C_0 \max\left\{\overline{R}_S^{(\nu+2)/(\nu+4)}, \overline{R}_S\right\} + \sqrt{V_{S,T}} + 4|S^c|/\nu. \tag{59}$$

where $C_0 = \sqrt{2/\pi} + 1$, $S = \{j : \sqrt{\nu} n\theta_j \geq c_0\}$ for some $c_0 > 0$ (e.g. $c_0 = 1$), and

$$\begin{aligned}
V_{S,T} &= \sum_{j\in S} \frac{1-\theta_j|S|}{\nu^2 n\theta_j} + \frac{1+8/\nu}{n^2}, \\
\overline{R}_S &= \frac{e\sqrt{8}}{\nu}\sum_{j\in S}\frac{\sqrt{n\theta_j}+1/2}{\sqrt{\nu}n\theta_j} + \frac{e\sqrt{8}}{n^{1/2}}\left(1 + \frac{1}{\nu} + \frac{1}{\nu^{3/2}} + \frac{1+1/\nu}{\sqrt{n\nu}}\right).
\end{aligned}$$

3 Log likelihood ratio and Hellinger statistic

The chi-squared and normal approximations are essentially equivalent for large contingency tables because the standardized chi-squared distribution is approximately Gaussian for large degrees-of-freedom. As classical Wilks' theorem has provided the chi-squared approximation of the log likelihood ratio when $d = O(1)$, we now focus on the normal approximation of the log likelihood ratio and Hellinger statistic when $d \to \infty$.

3.1 Log likelihood ratio goodness-of-fit test

Let $\lambda_j = n\theta_j$. By definition, the log likelihood ratio is given by

$$G = \sum_{j=1}^{d} f_{j,G}(X_j) = \sum_{j=1}^{d} 2X_j \log\big(X_j/\lambda_j\big), \tag{60}$$

where $f_{j,G}(x) = 2x\log(x/\lambda_j)$. Define

$$\sigma_{j,G} = (n\lambda_j)^{1/2}\sum_{k=1}^{d}\theta_k(\beta_j - \beta_k), \quad \sigma_G^2 = \sum_{j=1}^{d}\sigma_{j,G}^2, \quad V_G' = \sigma_G^2/n + \sum_{j=1}^{d} 1 \wedge \lambda_j^2, \tag{61}$$

where $\beta_j = \mathbb{E}[f_{j,G}(N_j+1) - f_{j,G}(N_j)]$ with $N_j \sim \mathrm{Pois}(\lambda_j)$ for $j \in [d]$. Similar to the σ^2 in (33), σ_G^2/n can be viewed as the variance of the linear part of the log likelihood ratio G. The following theorem is parallel to Corollary 2.9 for the asymptotic normality of G, i.e.,

$$\big(G - \mathbb{E}[G]\big)/\mathrm{Var}^{1/2}(G) \xrightarrow{\mathrm{D}} \mathcal{N}(0,1). \tag{62}$$

Theorem 3.1. *Let* $(X_1, \ldots, X_d) \sim \text{multinomial}(n, \theta_1, \ldots, \theta_d)$*, where* $d = d_n \geq 2$ *and* $\theta_j = \theta_{j,n}$ *for* $j \in [d]$ *are allowed to depend on n. Let* $V_G = \text{Var}(G)$ *be the variance of G and* V'_G *be as in* (61)*.*

(a) When $V_G \to \infty$*,* (62) *holds if and only if*

$$\frac{1}{d} + \frac{1}{V'_G} \max_{1 \leq j \leq d} \frac{\sigma^2_{j,G}/n}{1 + \lambda_j} + \frac{1}{V'_G} \sum_{j=1}^{d} \frac{\sigma^4_{j,G}/n^2}{\sigma^2_{j,G}/n + V'_G \lambda_j} \to 0. \tag{63}$$

(b) When $V_G \not\to \infty$ *and* $V_G \not\to 0$*,* (62) *holds if and only if*

$$\frac{1}{d} + \sum_{j=1}^{d} (1 - \theta_j)(1 \wedge \lambda_j^2) + \frac{1}{V'_G} \sum_{j=1}^{d} \frac{\sigma^4_{j,G}/n^2}{\sigma^2_{j,G}/n + V'_G \lambda_j} \to 0. \tag{64}$$

(c) When $V_G \to 0$ *and* $\limsup_{n\to\infty} \max_{1 \leq j \leq d} \theta_{j,n} < 1$*,* (62) *holds if and only if* $\sigma'^2_G > 0$ *and*

$$\frac{1}{d} + \sum_{j=1}^{d} \lambda_j^2 + \frac{1}{\sigma'^2_G} \sum_{j=1}^{d} \frac{\sigma'^4_{j,G}}{\sigma'^2_{j,G} + \sigma'^2_G \lambda_j} \to 0, \tag{65}$$

where $\sigma'_{j,G} = (n\lambda_j)^{1/2} \sum_{k=1}^{d} \theta_k \log(\theta_j/\theta_k)$ *and* $\sigma'^2_G = \sum_{j=1}^{d} \sigma'^2_{j,G}$*.*

Applying Steck's argument (Steck, 1955) for a central limit theory of conditional distributions, Morris (1975, Theorem 5.2) provided central limit theorems for Pearson's χ^2 and the log likelihood ratio allowing $d \to \infty$, under sufficient conditions that $\min_{1 \leq j \leq d} \lambda_j \geq \epsilon$ for some fixed $\epsilon > 0$ and $\max_{1 \leq j \leq d} \theta_j \to 0$. As we will see in Examples 3.4 and 3.5 in Section 3.3, neither of these two conditions are necessary.

3.2 Hellinger distance test for goodness of fit

Another closely related and important statistic is the Hellinger statistic

$$H = \sum_{j=1}^{d} f_{j,H}(X_j) = \sum_{j=1}^{d} 4(X_j^{1/2} - \lambda_j^{1/2})^2, \tag{66}$$

where $f_{j,H}(x) = 4(x^{1/2} - \lambda_j^{1/2})^2$. Similarly, define

$$\sigma_{j,H} = (n\lambda_j)^{1/2} \sum_{k=1}^{d} \theta_k (\gamma_j - \gamma_k), \quad \sigma^2_H = \sum_{j=1}^{d} \sigma^2_{j,H}, \quad V'_H = \sigma^2_H/n + \sum_{j=1}^{d} 1 \wedge \lambda_j^3, \tag{67}$$

where $\gamma_j = \mathbb{E}[f_{j,H}(N_j + 1) - f_{j,H}(N_j)]$ with $N_j \sim \text{Pois}(\lambda_j)$ for $j \in [d]$. Likewise, σ^2_H/n can be viewed as the variance of the linear part of the Hellinger statistic H. The following theorem provides necessary and sufficient conditions for the asymptotic normality of H, i.e.,

$$\big(H - \mathbb{E}[H]\big)/\text{Var}^{1/2}(H) \xrightarrow{\text{D}} \mathcal{N}(0, 1). \tag{68}$$

Theorem 3.2. *Let* $(X_1, \ldots, X_d) \sim \text{multinomial}(n, \theta_1, \ldots, \theta_d)$*, where* $d = d_n \geq 2$ *and* $\theta_j = \theta_{j,n}$ *for* $j \in [d]$ *are allowed to depend on n. Let* $V_H = \text{Var}(H)$ *be the variance of H and* V'_H *be as in* (67)*.*

(a) When $V_H \to \infty$*,* (68) *holds if and only if*

$$\frac{1}{d} + \frac{1}{V'_H} \max_{1 \leq j \leq d} \frac{\sigma^2_{j,H}/n}{1 + \lambda_j} + \frac{1}{V'_H} \sum_{j=1}^{d} \frac{\sigma^4_{j,H}/n^2}{\sigma^2_{j,H}/n + V'_H \lambda_j} \to 0. \tag{69}$$

(b) When $V_H \not\to \infty$ and $V_H \not\to 0$, (68) holds if and only if

$$\frac{1}{d} + \sum_{j=1}^{d} (1-\theta_j)(1 \wedge \lambda_j^4) + \frac{1}{V'_H} \sum_{j=1}^{d} \frac{\sigma_{j,H}^4/n^2}{\sigma_{j,H}^2/n + V'_H \lambda_j} \to 0. \tag{70}$$

(c) When $V_H \to 0$ and $\limsup_{n\to\infty} \max_{1\le j\le d} \theta_{j,n} < 1$, (68) holds if and only if

$$\frac{1}{d} + \sum_{j=1}^{d} \lambda_j^3 + \sum_{j=1}^{d} \frac{\lambda_j^3}{V''_H} \wedge \frac{\lambda_j^4}{V''^2_H} + \frac{1}{V''_H} \sum_{j=1}^{d} \frac{\sigma'^4_{j,H}/n^2}{\sigma'^2_{j,H}/n + V''_H \lambda_j} \to 0, \tag{71}$$

where $\sigma'_{j,H} = (n\lambda_j)^{1/2} \sum_{k=1}^{d} \theta_k(\eta_j - \eta_k)$ with $\eta_j = \lambda_j^{1/2}(2\lambda_j - 2 - \sqrt{2})$ for $j \in [d]$ and $V''_H = \sum_{j=1}^{d} \{\sigma'^2_{j,H}/n + \lambda_j^3\}$.

In Theorems 2.7, 3.1 and 3.2, the necessary and sufficient conditions involving σ_j^4, $\sigma_{j,G}^4$ or $\sigma_{j,H}^4$ are derived from Lindeberg's conditions for the linear and non-linear components of certain Poissonization of these statistics.

3.3 Examples

When $\theta_j = 1/d$ for $j \in [d]$, the multinomial model is called *equiprobable*.

Example 3.3 (Equiprobable)**.** *For equiprobable models, Pearson's χ^2, the log likelihood ratio and Hellinger statistics have approximately the χ^2_{d-1} distribution if and only if $n^2/d \to \infty$, and they are asymptotically normal if and only if $d \wedge (n^2/d) \to \infty$. In particular, for $n \ll d \ll n^2$, they are asymptotically normal even if $\lambda_j = n/d \to 0$.*

Pearson's χ^2, the log likelihood ratio and the Hellinger statistics can be asymptotically normal even when there is a large cell, e.g., $\theta_{1,n} \to 1$. Thus, the condition $\limsup_{n\to\infty} \max_{1\le j\le d} \theta_{j,n} < 1$ is *not* necessary in Theorem 3.1 (c) and Theorem 3.2 (c).

Example 3.4 (One large cell)**.** *Consider $(X_1, \dots, X_d) \sim \text{multinomial}(n, \theta_1, \dots, \theta_d)$ with*

$$\theta_1 = 1 - n^{-\alpha}, \quad \theta_2 = \cdots = \theta_d = n^{-\alpha}/(d-1),$$

where $\alpha \in (0,1)$ not depending on n and $d = d_n \to \infty$. In this case, for $d \ll n^2$, Pearson's χ^2 is asymptotically normal by Corollary 2.9. Moreover, for $d \gg n^{2(1-\alpha)}$, the log likelihood ratio G and the Hellinger statistic H are asymptotically normal by Theorem 3.1 (a) and Theorem 3.2 (a) respectively.

Moreover, these statistics can be asymptotically normal even when $n^2/d + \max_{1\le j\le d} \lambda_j \to 0$.

Example 3.5 (ϵ-Perturbation)**.** *Consider $(X_1, \dots, X_d) \sim \text{multinomial}(n, \theta_1, \dots, \theta_d)$ with*

$$\theta_1 = \cdots = \theta_{d/2} = (1+\epsilon)/d, \quad \theta_{d/2+1} = \cdots = \theta_d = (1-\epsilon)/d,$$

where $\epsilon = \epsilon_n \in (-1,1)\backslash\{0\}$ and $d = d_n \gg n^2$ is an even integer. In this case, for $\epsilon = 1/2$ in particular, Pearson's χ^2 statistic, the log likelihood ratio G and the Hellinger statistic H are asymptotically normal by Corollary 2.9, Theorem 3.1 (a) and Theorem 3.2 (a) respectively.

In Example 3.5, the log likelihood ratio G is asymptotically normal when $V'_G \to [0, \infty]$ in various cases. For instance, $V'_G \to \infty$ when ϵ is a constant not depending on n, $V'_G \asymp 1$ when $\epsilon = \tanh(n^{-1/2}/2)$, while $V'_G \to 0$ when $\epsilon = \tanh(n^{-1/2-\delta}/2)$ for a small constant $\delta > 0$ not depending on n.

For each statistic $T \in \{\chi^2, G, H\}$, we plot $\mathbb{E}[T]$ versus $\text{Var}(T)/2$ in the logarithmic scale for multinomial models in above examples. We set $d = \lceil n^{1.2} \rceil$ in all three examples, $\alpha = 1/2$ in Example 3.4, and $\epsilon = 1/2$ in Example 3.5. It can be seen from Figure 1 that $\text{Var}(T)/2$ is significantly different from $\mathbb{E}[T]$ in most cases. This supports the use of the degrees-of-freedom adjusted chi-squared approximation or the normal approximation to match both the mean and variance of these statistics.

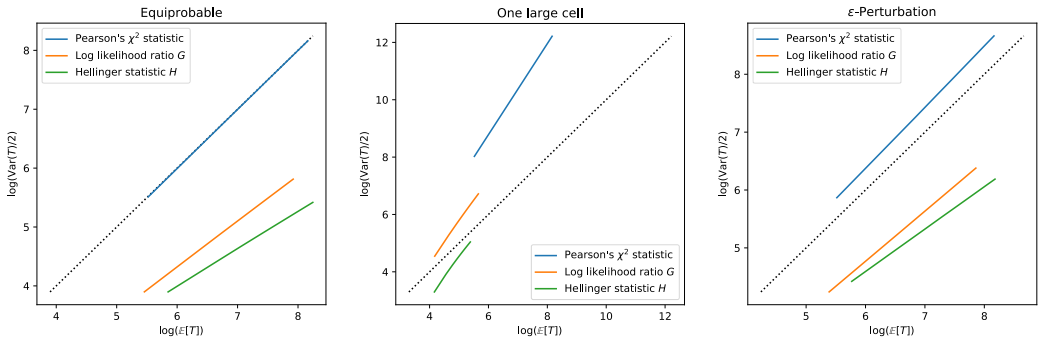


Figure 1: Mean versus half variance for χ^2, G and H in Examples 3.3, 3.4 and 3.5

4 Simulation results

4.1 Type I error rates comparison for tests of independence

We compare the performance in terms of the type I error rates of five approximation schemes, namely the chi-squared approximation for Pearson's χ^2_{indep} with degrees of freedom $\nu = (d_1-1)(d_2-1)$ and the adjusted degrees of freedom $\overline{\nu}$ in (17), the chi-squared approximation for the cross-sample chi-squared statistic $\chi^2_{\text{CS-indep}}$ with degrees of freedom $\nu = (d_1-1)(d_2-1)$ and the adjusted degrees of freedom ν^* and $\overline{\nu}_{\text{CS}}$ in (10) and (9).

In our simulations, we consider square contingency tables with $d_1 = d_2 = d$ and cell probabilities $p_j q_k$ generated in the following three cases:

- Case 1: For $j \in [d]$, sample i.i.d. $\widetilde{p}_j \sim \text{beta}(2,2)$ from a Beta distribution, and set the marginal row probability $p_j = \widetilde{p}_j / \sum_{j=1}^d \widetilde{p}_j$; Generate the marginal column probability $q_{1:d}$ the same way.
- Case 2: For $j \in [d]$, sample i.i.d. $\widetilde{p}_j \sim |\mathcal{N}(0,1)|$ from a folded normal distribution, and set the marginal row probability $p_j = \widetilde{p}_j / \sum_{j=1}^d \widetilde{p}_j$; Generate $q_{1:d}$ the same way.
- Case 3: Let $L = \lceil 0.7d \rceil$, sample i.i.d. $\widetilde{p}_j \sim \text{beta}(2,2)$ for $1 \le j \le L$, set $\widetilde{p}_j = 0.6^{j+1-L}$ for $L < j \le d$, and $p_j = \widetilde{p}_j / \sum_{j=1}^d \widetilde{p}_j$ for $j \in [d]$; Generate $q_{1:d}$ the same way.

Table 1: *Type I Error Rates, Case 1*

	df		χ^2_{indep}		$\chi^2_{\text{CS-indep}}$			χ^2_{indep}		$\chi^2_{\text{CS-indep}}$		
$(d, n/d)$	$\overline{\nu}$	$\overline{\nu}_{\text{CS}}$	χ^2_ν	$\chi^2_{\overline{\nu}}$	χ^2_ν	$\chi^2_{\nu^*}$	$\chi^2_{\overline{\nu}_{\text{CS}}}$	χ^2_ν	$\chi^2_{\overline{\nu}}$	χ^2_ν	$\chi^2_{\nu^*}$	$\chi^2_{\overline{\nu}_{\text{CS}}}$
	$\text{df} \times 10^{-4}$		$\alpha = 0.05$, $\hat{\alpha} \times 10^2$					$\alpha = 0.1$, $\hat{\alpha} \times 10$				
$(50, 5)$	0.440	0.109	0.98	6.11	1.52	**5.86**	5.88	0.15	1.05	0.39	**1.04**	1.04
$(70, 5)$	1.123	0.245	0.49	6.09	1.50	**5.91**	5.92	0.07	1.05	0.39	**1.05**	1.05
$(90, 5)$	2.289	0.35**6**	0.26	6.02	1.48	**5.95**	5.96	0.04	1.05	0.39	**1.05**	1.05
$(200, 10)$	16.60	2.73**5**	2.45	5.78	3.04	**5.63**	**5.63**	0.32	**0.99**	0.67	1.03	1.03
$(200, 50)$	6.241	3.714	8.00	**5.15**	4.63	5.17	5.17	1.28	0.97	0.94	**1.01**	**1.01**
$(400, 10)$	122.2	10.98	0.85	5.86	3.06	**5.63**	**5.63**	0.11	**1.02**	0.67	1.03	1.03
$(400, 50)$	35.24	14.93	9.85	5.17	4.62	**5.14**	**5.14**	1.44	0.97	0.93	**1.00**	**1.00**
$(600, 10)$	400.5	24.75	0.28	5.79	3.04	**5.63**	**5.63**	0.03	**1.02**	0.67	1.03	1.03
$(600, 50)$	102.3	33.64	11.0	5.25	4.63	**5.16**	**5.16**	1.52	0.97	0.94	**1.01**	**1.01**

Compared with Cases 1 and 2, the cell probabilities in Case 3 contain more very small elements, producing more sparse contingency tables.

In each of the three cases, we take 9 combinations of d and n and repeat the experiment independently for 10^6 times. In each experiment, we sample a random matrix $(X_{j,k})_{d\times d} \sim \text{multinomial}(n, (p_j q_k)_{d\times d})$, and compute the test statistics χ^2_{indep} and $\chi^2_{\text{CS-indep}}$. Based on the 10^6 realizations of these test statistics and for target test sizes $\alpha = 0.05$ and $\alpha = 0.1$, the proportions of false rejections are calculated and recorded in Tables 1, 2 and 3 for the three cases.

Table 2: *Type I Error Rates, Case 2*

	df		χ^2_{indep}		$\chi^2_{\text{CS-indep}}$			χ^2_{indep}		$\chi^2_{\text{CS-indep}}$		
$(d, n/d)$	$\overline{\nu}$	$\overline{\nu}_{\text{CS}}$	χ^2_{ν}	$\chi^2_{\overline{\nu}}$	χ^2_{ν}	$\chi^2_{\nu^*}$	$\chi^2_{\overline{\nu}_{\text{CS}}}$	χ^2_{ν}	$\chi^2_{\overline{\nu}}$	χ^2_{ν}	$\chi^2_{\nu^*}$	$\chi^2_{\overline{\nu}_{\text{CS}}}$
	$\text{df} \times 10^{-4}$		$\alpha = 0.05, \quad \hat{\alpha} \times 10^2$					$\alpha = 0.1, \quad \hat{\alpha} \times 10$				
$(50, 5)$	0.667	0.083	0.04	6.26	0.82	**5.81**	5.83	0.01	1.07	0.25	**1.03**	1.04
$(70, 5)$	1.767	0.16**4**	0.01	6.15	0.82	**5.88**	5.89	0.00	1.06	0.25	**1.03**	1.04
$(90, 5)$	3.678	0.27**3**	0.00	6.09	0.80	**5.88**	5.89	0.00	1.07	0.25	**1.04**	1.04
$(200, 10)$	47.90	2.183	0.03	6.23	2.02	**5.65**	**5.65**	0.00	1.05	0.50	**1.02**	**1.02**
$(200, 50)$	36.80	3.403	5.88	**5.11**	4.06	5.22	5.22	0.69	0.85	0.85	**1.01**	**1.01**
$(400, 10)$	375.1	8.76**9**	0.00	5.96	2.02	**5.70**	**5.70**	0.00	1.05	0.50	**1.03**	**1.03**
$(400, 50)$	281.8	13.68	3.83	5.34	4.14	**5.33**	**5.33**	0.43	0.89	0.86	**1.01**	**1.01**
$(600, 10)$	1257	19.76	0.00	5.84	2.03	**5.68**	**5.68**	0.00	1.04	0.49	**1.03**	**1.03**
$(600, 50)$	936.9	30.82	2.52	5.49	4.11	**5.25**	**5.25**	0.28	0.91	0.85	**1.01**	**1.01**

In each row in Tables 1, 2 and 3, the entry closest to the target value α is marked in bold. It can be observed that the degrees of freedom adjusted $\chi^2_{\nu^*}$ approximation of the cross-sample chi-squared statistic $\chi^2_{\text{CS-indep}}$ is the most robust scheme with the best overall size accuracy. For the classic χ^2_{indep} statistic, the degrees of freedom adjusted $\chi^2_{\overline{\nu}}$ approximation has much better size accuracy than the traditional χ^2_{ν} approximation.

Table 3: *Type I Error Rates, Case 3*

	df		χ^2_{indep}		$\chi^2_{\text{CS-indep}}$			χ^2_{indep}		$\chi^2_{\text{CS-indep}}$		
$(d, n/d)$	$\overline{\nu}$	$\overline{\nu}_{\text{CS}}$	χ^2_{ν}	$\chi^2_{\overline{\nu}}$	χ^2_{ν}	$\chi^2_{\nu^*}$	$\chi^2_{\overline{\nu}_{\text{CS}}}$	χ^2_{ν}	$\chi^2_{\overline{\nu}}$	χ^2_{ν}	$\chi^2_{\nu^*}$	$\chi^2_{\overline{\nu}_{\text{CS}}}$
	$\text{df} \times 10^{-4}$		$\alpha = 0.05, \quad \hat{\alpha} \times 10^2$					$\alpha = 0.1, \quad \hat{\alpha} \times 10$				
$(50, 5)$	0.354	0.079	0.00	6.31	0.62	**5.67**	5.68	0.00	1.03	0.21	**1.03**	1.03
$(70, 5)$	0.746	0.151	0.00	6.29	0.54	**5.71**	5.72	0.00	1.04	0.19	**1.03**	1.03
$(90, 5)$	1.335	0.24**5**	0.00	6.29	0.49	**5.75**	5.75	0.00	1.04	0.18	**1.04**	1.04
$(200, 10)$	8.438	1.599	0.00	5.57	0.79	**5.48**	**5.48**	0.00	0.93	0.27	**1.03**	**1.03**
$(200, 50)$	10.23	2.045	0.00	3.53	1.27	**5.12**	**5.12**	0.00	0.61	0.39	**1.00**	**1.00**
$(400, 10)$	46.22	6.24**9**	0.00	5.66	0.71	**5.51**	**5.51**	0.00	0.97	0.26	**1.03**	**1.03**
$(400, 50)$	34.20	7.837	0.00	3.95	1.07	**5.14**	**5.14**	0.00	0.71	0.35	**1.01**	**1.01**
$(600, 10)$	133.9	13.95	0.00	5.71	0.69	**5.50**	**5.50**	0.00	**0.98**	0.25	1.03	1.03
$(600, 50)$	75.41	17.38	0.00	4.25	1.01	**5.16**	**5.16**	0.00	0.77	0.34	**1.01**	**1.01**

In our simulation experiments, ϵ_n is always negative and much smaller than ν^* in absolute value, so that the adjusted degrees of freedom ν^* in (9) provides a highly accurate approximation of the half conditional

variance $\overline{\nu}_{\text{CS}}$. In fact, their numerical values are virtually identical in the simulations except for a few cases, where the difference is only by one in the last digit marked in bold in the $\overline{\nu}_{\text{CS}}$-column of the tables.

As the degrees of freedom ν, $\overline{\nu}$ and $\overline{\nu}_{\text{CS}} \approx \nu^*$ are significantly different and diverge with d, the approximation schemes (12) and (20) matching both the conditional mean and variance of the test statistics are more robust and have better size accuracy throughout the low- and high-dimensional regimes as demonstrated by the numerical results. Since $\epsilon_n \leq 0$ in most cases and $\overline{\nu}_{\text{CS}} \leq \nu^*$, the $\chi^2_{\nu^*}$ approximation is slightly less conservative than the $\chi^2_{\overline{\nu}}$ approximation. Therefore, our simulations support the use of degrees of freedom adjusted $\chi^2_{\nu^*}$ approximation of the cross-sample $\chi^2_{\text{CS-indep}}$ statistic for the test of independence in large and sparse two-way contingency tables.

4.2 Power analysis for the tests of independence

To validate Theorem 2.6, we compare the empirical powers to the theoretical powers implied by Theorem 2.6. Two types of null cell probabilities $\{\kappa_{j,k} = p_j q_k\}_{j,k=1,\ldots,d}$, described below, are considered. The true cell probabilities $\{\theta_{j,k}\}_{j,k=1,\ldots,d}$ are drawn as follows.

- Case 1: Let $p'_j = q'_k = 1/d$ for all $j, k \in [d]$. Set for some $0 \leq a < 1$

$$\tilde{\theta}_{j,k} = \Big(1 + (-1)^{j+k-1} a\Big) p'_j q'_k.$$

  Finally, obtain $p_j = \sum_k \tilde{\theta}_{j,k}$, $q_k = \sum_j \tilde{\theta}_{j,k}$.

- Case 2: The $\{p'_j, q'_k\}_{j,k\in[d]}$ are generated in the same way as Case 1 in Section 4.1. Set for some $0 \leq a < 1$

$$\tilde{\theta}_{j,k} = \begin{cases} (1+a) p'_j q'_k & \text{if } j = k \\ (1-a) p'_j q'_k & \text{otherwise} \end{cases}$$

  Finally, normalize $\theta_{j,k} = \tilde{\theta}_{j,k} \big/ \big(\sum_{j,k} \tilde{\theta}_{j,k}\big)$ and obtain $p_j = \sum_k \theta_{j,k}$, $q_k = \sum_j \theta_{j,k}$.

The quantity $a > 1$ above measures the deviation of $\{\kappa_{j,k}\}$ from the true $\{\theta_{j,k}\}$. The larger a is, the larger is the difference between the null and alternative approximating distributions, and hence the higher is the power.

We vary (d, n, a) for each type of κ, and compare the empirical and theoretical powers in every combination $(\kappa_{1:d}, d, n, a)$. Given a $(\kappa_{1:d}, d, n, a)$, the empirical powers are computed by repeating the following steps independently for 1×10^4 times: sample the data vector $\{X_{j,k}\}$ from multinomial$(n, \theta_{j,k})$; compute the test statistic $\chi^2_{\text{CS-indep}}$; choose the cut-off values for $\chi^2_{\text{CS-indep}}$ at level α to be $\Phi^{-1}_{\nu^*}(1-\alpha) - \nu^*$ with $\Phi^{-1}_{\omega}(1-\alpha)$ being the $1-\alpha$ quantile of a chi-squared distribution with degrees of freedom ω; compare the test statistic $\chi^2_{\text{CS-indep}}$ with the cutoff value and either reject or accept the H_0. The distributions of the empirical powers at various a's are exhibited as black circles with error bars in Figure **??**. The theoretical powers are computed as follows. Theorem 2.6 states that when $\{\kappa_{j,k}\} \neq \{\theta_{j,k}\}$, the variance of $\chi^2_{\text{CS-indep}}$ matches the variance of a non-central chi-squared distribution $\chi^2_{\omega}(\tau)$ with degrees of freedom ω and noncentrality parameter τ, that is

$$V_{\text{CS-indep}} \approx 2\omega + 4\tau,$$

where $V_{\text{CS-indep}}$ is as in (27). We choose ω and τ to be

$$\omega = \min\{(d_1 - 1)(d_2 - 1), V_{\text{CS-indep}}/2\}, \quad \tau = V_{\text{CS-indep}}/4 - \omega/2.$$

The theoretical power of the cross-sample version test is

$$\begin{aligned}&\mathbb{P}\Big\{\chi^2_\omega(\tau) - (\omega+\tau) + \mu_{\text{CS-indep}} \geq \Phi^{-1}_{\nu^*}(1-\alpha) - \nu^*\Big\}\\ =&1 - \Phi_{\omega,\tau}\Big(\Phi^{-1}_{\nu^*}(1-\alpha) - \nu^* + (\omega+\tau) - \mu_{\text{CS-indep}}\Big),\end{aligned}$$

where $\mu_{\text{CS-indep}}$ is defined in (24), and $\Phi_{\omega,\tau}(\cdot)$ is the CDF of the non-central chi-square distribution $\chi^2_\omega(\tau)$.

Table 4: *Case 2, d=50, n=500*

theo_power	empi_power	theo_mean	emp_mean	theo_var_1	theo_var_2	emp_var
0.0868	0.0569	0.0000	-0.1743	4792.3960	4792.3960	3278.6538
0.0883	0.0544	0.5862	-0.2793	4800.6382	4802.527	3176.1545
0.0952	0.0617	3.1366	2.8063	4850.9603	4852.665	3353.1650
0.1160	0.0806	9.9226	9.5816	5010.1874	5004.261	3386.3866
0.1778	0.1327	26.6078	28.0681	5475.5768	5437.091	3656.8658
0.2140	0.1725	35.0971	36.6623	5739.0377	5679.849	3875.5176
0.2654	0.2203	46.3378	47.0818	6108.9494	6019.390	4059.1842
0.3378	0.3022	61.4026	64.0896	6636.3484	6501.882	4432.7396
0.4365	0.4199	81.9047	84.1355	7402.7595	7201.054	4842.0931
0.5629	0.5756	110.3493	113.5114	8543.0195	8238.858	5554.4964
0.7070	0.7673	150.7795	155.2109	10289.5114	9825.534	6394.6123
0.8431	0.9204	210.0305	217.7004	13063.0270	12341.958	8313.2097
0.9400	0.9891	300.3450	311.1163	17673.0000	16521.082	10866.8323
0.9862	0.9998	445.3187	461.5734	25798.7279	23883.356	15833.2849
0.9986	1.0000	695.0446	725.3056	41280.3646	37900.191	26479.4432

Table 5: *Case 1, d=50, n=1500*

theo_power	empi_power	theo_mean	emp_mean	theo_var	emp_var
0.0556	0.0496	0.0000	0.5065	4798.7987	4445.5975
0.0596	0.0549	2.2198	1.9191	4817.3315	4518.8960
0.0794	0.0743	11.9075	11.8614	4914.9108	4558.8649
0.1548	0.1485	37.7941	38.5428	5238.8198	4959.1122
0.4483	0.4501	101.8306	102.8487	6258.0830	5829.2130
0.5779	0.5731	128.2477	128.4753	6742.0577	6360.5395
0.7184	0.7218	161.5970	162.4110	7395.0197	6861.9399
0.8458	0.8587	204.0362	203.5686	8285.2528	7512.4663
0.9358	0.9529	258.5743	258.4397	9514.2919	8517.8534
0.9815	0.9879	329.4966	329.0423	11236.5718	10421.0512
0.9967	0.9985	423.0528	423.3693	13693.1374	12328.0230
0.9997	0.9999	548.6149	549.7539	17271.5813	15607.5997
1.0000	1.0000	720.7026	724.8403	22617.0429	20602.7122
1.0000	1.0000	962.7005	961.7051	30847.7117	27922.1457
1.0000	1.0000	1314.0816	1312.0225	43996.8987	39936.1803

We also compare the empirical powers of the five approximation schemes in Tables 1-3 and the U-statistic permutation test statistic (USP) proposed in (Berrett et al., 2021). The comparisons are made under four different cases. In each case, the empirical power is the average over 1×10^4 repetitions.

Table 6: *Case 1, d=150, n=4000*

theo_power	empi_power	theo_mean	emp_mean	theo_var	emp_var
0.0568	0.0486	0.0000	-1.8185	44390.8995	40259.7422
0.0579	0.0539	1.9944	4.1981	44424.6904	41490.0161
0.0632	0.0589	10.8044	11.4710	44589.9672	42183.1068
0.0796	0.0700	34.7646	34.2053	45102.6397	41156.0951
0.1345	0.1277	95.5743	97.3451	46636.0229	42684.6620
0.1635	0.1513	121.1346	120.9197	47351.4649	43429.8278
0.2052	0.1936	153.7311	153.8870	48312.7765	44404.7389
0.2661	0.2515	195.6918	193.4054	49621.4904	45520.1817
0.3544	0.3463	250.3283	252.1491	51431.3328	47474.2287
0.4782	0.4802	322.4641	325.2503	53981.5870	49016.7609
0.6371	0.6355	419.3199	418.7079	57656.8509	53574.5198
0.8067	0.8178	552.0591	553.7798	63098.9542	57807.1029
0.9353	0.9436	738.6123	738.7509	71427.8576	66386.4465
0.9901	0.9954	1009.1435	1010.3837	84705.8314	75911.8092
0.9996	1.0000	1417.3717	1416.5372	106989.9569	95678.9841

- Case 1: The cell probabilities are of the form

$$\theta_{j,k} = \frac{2^{-j-k}}{(1-2^{-d})^2} + a \cdot (\mathbb{1}_{j=k=1} + \mathbb{1}_{j=k=2}) - a \cdot (\mathbb{1}_{j=1,k=2} + \mathbb{1}_{j=2,k=1}),$$

  where $1 \le j, k \le d = 6$ and $a \ge 0$. The sample size is $n = 100$.

- Case 2: The cell probabilities are of the form

$$\theta_{j,k} = \frac{1}{d^2} + (-1)^{j+k-1} a,$$

  where $1 \le j, k \le d = 8$ and $a \ge 0$. The sample size is $n = 50$.

- Case 3: The $\{p'_j, q'_k\}_{j,k\in[d]}$ are generated in the same way as Case 1 in Section 4.1. Set for some $0 \le a < 1$

$$\tilde{\theta}_{j,k} = \begin{cases} (1+a)p'_j q'_k & \text{if } j = k \\ (1-a)p'_j q'_k & \text{otherwise} \end{cases}$$

  Finally, normalize $\theta_{j,k} = \tilde{\theta}_{j,k} \big/ \big(\sum_{j,k} \tilde{\theta}_{j,k}\big)$ and obtain $p_j = \sum_k \theta_{j,k}$, $q_k = \sum_j \theta_{j,k}$.

- Case 4: The $\{p'_j, q'_k\}_{j,k\in[d]}$ are generated in the same way as Case 3 in Section 4.1. Let $\mathcal{B} := \{(j,k) : p'_j q'_k \ge Q\}$, where Q is the 80% quantile of all $\{p'_j q'_k\}_{j,k}$. The cell probabilities are of the form

$$\tilde{\theta}_{j,k} = \Big(1 + (-1)^{j+k-1} a \cdot \mathbb{1}_{(j,k)\in\mathcal{B}}\Big) p'_j q'_k$$

  for some $0 \le a < 1$. Finally, normalize $\theta_{j,k} = \tilde{\theta}_{j,k} \big/ \big(\sum_{j,k} \tilde{\theta}_{j,k}\big)$.

Case 1 and 2 are the same as in Section 7.2 of Berrett et al. (2021). In Case 3, the departure from independence occurs in all cells, and is proportional to the cell probability. In Case 4, the alternative departs from independence in large-probability cells but not in small-probability cells.

As shown in Figure 2, the schemes "$\chi^2_{\text{CS-indep}}$ to approximate $\chi^2_{\nu^*}$" and "$\chi^2_{\text{CS-indep}}$ to approximate $\chi^2_{\bar{\nu}_{\text{CS}}}$" have comparable performances with the rate-optimal USP test in high-dimensional settings.

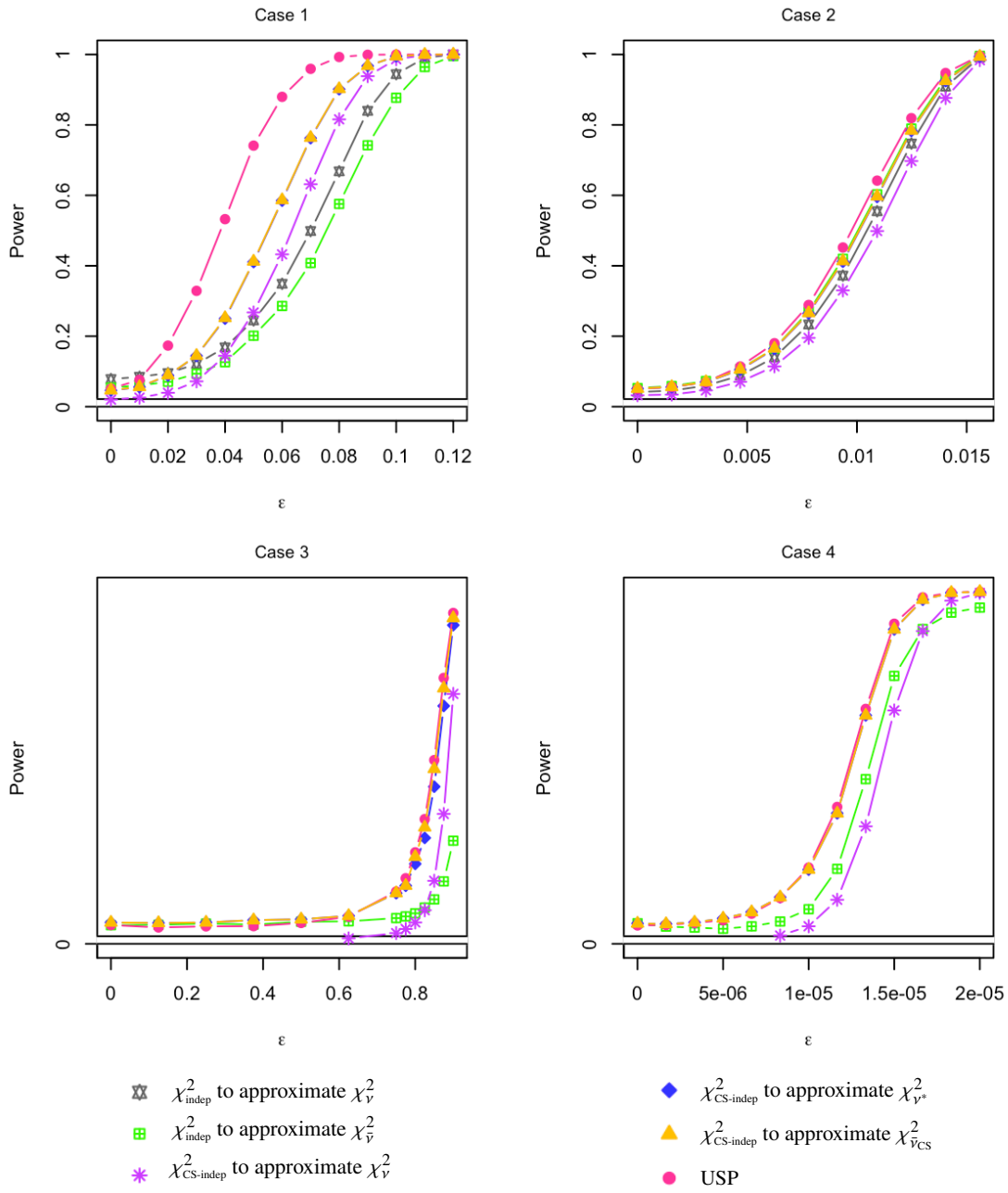


Figure 2: Comparison of the empirical powers of six tests. Each point represents average of 1×10^4 independent repetitions.

Table 7: *Case 2, d=50, n=1500*

theo_power	empi_power	theo_mean	emp_mean	theo_var	emp_var
0.0591	0.0511	0.0000	-0.6407	4798.7987	4283.1085
0.0630	0.0573	2.1266	2.4383	4815.2660	4347.3311
0.0827	0.0731	11.3814	12.0832	4907.9436	4433.8724
0.1554	0.1440	36.0126	35.8820	5216.5223	4668.7209
0.4325	0.4289	96.6033	98.2384	6179.8611	5534.9646
0.5562	0.5554	121.4997	122.5079	6634.2536	5990.5229
0.6934	0.6966	152.8636	153.3150	7245.0424	6462.5908
0.8232	0.8355	192.6833	191.1044	8074.3666	7065.0874
0.9207	0.9387	243.7220	244.6568	9214.1137	8187.1223
0.9747	0.9853	309.8986	309.7318	10803.2101	9509.6745
0.9948	0.9985	396.9038	397.9533	13057.0975	11414.7537
0.9994	0.9999	513.2290	513.1553	16319.7931	14447.3459
1.0000	1.0000	671.9567	671.7263	21159.6580	18434.6113
1.0000	1.0000	894.0259	896.5522	28554.1417	25308.1391
1.0000	1.0000	1214.5329	1215.7190	40266.7432	34882.2624

4.3 Type I error rates comparison for the goodness-of-fit tests

Under the same cases as in Xu et al. (2019), simulations are conducted to compare the size accuracy of the chi-squared and normal approximations of Pearson's χ^2, its cross-sample version χ^2_{CS}, the log likelihood ratio and Hellinger statistics. We consider two cases, in which the null distributions are respectively,

- Case 1 (Example 3.5 with $\epsilon = 1/2$): Set $\theta_j = 1/(2d)$ for $1 \le j \le d/2$, and set $\theta_j = 3/(2d)$ for $d/2+1 \le j \le d$.
- Case 2: Set $\theta_j = 2^{-d/2}(d/2-1)!\big/\big((j-1)!(d/2-j)!\big)$ for $1 \le j \le d/2$, and set $\theta_j = 1/d$ for $d/2+1 \le j \le d$.

In each case, we let (n, d) vary in $\{40, 100, 400\}$ and repeat the experiment independently for 6×10^6 times. In each experiment, we sample $(X_1, \ldots, X_d) \sim \text{multinomial}(n, \theta_1, \ldots, \theta_d)$ and compute the test statistics χ^2, χ^2_{CS}, G and H under the null hypothesis. For target significance level $\alpha = 0.05$, type I error rates are calculated and recorded in Tables 9 and 10.

We compare the performances of the normal approximation, the chi-squared approximation with degrees of freedom matching the mean, and the centered chi-squared approximation with degrees of freedom matching the half variance. For simplicity, the two chi-squared approximation schemes are called "traditional" and "adjusted" respectively. For Pearson's χ^2, the log likelihood ratio G and the Hellinger statistic H, all three approximation schemes are considered. For the cross sample chi-squared statistic χ^2_{CS}, the half variance is $(d-1)(1-1/n)$, so that only the normal and χ^2_{d-1} approximations are considered. This gives totally 11 approximation schemes for the 4 test statistics. The three best results are marked in bold in each row.

In Table 9, where the cell probability θ_j's are relatively balanced in Case 1, the adjusted chi-squared approximation is the best overall performer, followed by the normal approximation. This is expected because the two better methods match test statistics in both the mean and variance. Among the four statistics, Pearson's χ^2 is over all the least stable and demonstrate a fatter right tail than the chi-squared distribution. This is probably caused by the singularities at small cell probability θ_j's in the denominators.

In Table 10, where the cell probabilities are unbalanced with some very small θ_j's in Case 2, the log likelihood ratio demonstrates most stable results overall, followed by the Hellinger statistic. Again, the adjusted chi-squared approximation performs the best, followed by the normal approximation.

Table 8: *Case 2, d=150, n=4000*

theo_power	empi_power	theo_mean	emp_mean	theo_var	emp_var
0.0608	0.0530	-0.0000	2.9718	44390.8995	39333.2587
0.0620	0.0568	1.9543	0.4743	44421.9310	39577.4328
0.0674	0.0605	10.5792	14.4407	44582.9187	39552.2174
0.0842	0.0720	34.0034	30.8264	45086.5525	40832.0124
0.1395	0.1219	93.3331	93.3404	46592.9960	40334.2526
0.1683	0.1580	118.2347	123.0909	47294.6934	41775.8814
0.2096	0.1976	149.9659	152.0386	48236.4061	43459.3135
0.2693	0.2497	190.7759	188.5374	49516.5749	44484.2637
0.3555	0.3389	243.8593	242.0095	51283.8702	44706.4035
0.4759	0.4718	313.8613	315.8478	53769.0492	47755.3680
0.6307	0.6348	407.7220	409.7099	57341.8061	50041.9130
0.7980	0.8107	536.1459	535.9399	62616.7991	56240.0845
0.9289	0.9400	716.2806	713.1279	70661.9108	62392.7853
0.9881	0.9947	976.8780	979.6849	83433.3907	72325.4295
0.9994	0.9999	1368.9406	1367.7441	104755.8693	92907.9815

Table 9: *Type I Error Rates* $\times 10^2$ *at* $\alpha = 0.05$*, Case 1*

	Pearson's χ^2			Cross-sample χ^2_{CS}		Log likelihood ratio G			Hellinger H		
(d, n)	$\mathcal{N}(0,1)$	χ^2_{d-1}	$\chi^2_{V/2}$	$\mathcal{N}(0,1)$	χ^2_{d-1}	$\mathcal{N}(0,1)$	$\chi^2_{\mu_G}$	$\chi^2_{V_G/2}$	$\mathcal{N}(0,1)$	$\chi^2_{\mu_H}$	$\chi^2_{V_H/2}$
(40, 40)	6.86	6.86	5.64	5.50	**5.50**	5.67	1.80	**4.51**	5.50	3.46	**4.58**
(40, 100)	6.30	5.77	5.29	6.24	**5.18**	5.91	4.17	**4.83**	5.99	8.97	**5.27**
(40, 400)	6.08	5.14	**5.02**	6.05	**4.98**	6.10	5.41	**5.06**	6.69	8.87	5.88
(100, 40)	7.04	9.27	**5.31**	6.30	6.30	**5.35**	0.24	4.16	**5.24**	0.06	4.07
(100, 100)	6.43	6.43	5.62	5.95	**5.10**	5.55	1.78	4.65	**5.34**	3.49	**4.70**
(100, 400)	5.90	5.46	**5.04**	5.74	**5.08**	5.61	**5.05**	4.90	5.69	10.60	5.21
(400, 40)	6.28	14.40	6.28	5.96	5.96	**5.63**	0.00	**4.30**	**5.34**	0.00	3.47
(400, 100)	6.39	9.99	6.39	5.57	**5.57**	**5.31**	0.04	4.40	**5.18**	0.00	4.20
(400, 400)	5.83	6.77	5.40	5.38	5.38	5.29	1.76	**4.82**	**5.17**	3.56	**4.83**

4.4 Power analysis for the tests of goodness-of-fit

We conduct some numerical experiments to verify the accuracy of the approximate power functions implied by Theorems 2.11 and 2.12. Two types of $\kappa_{1:d}$'s (null probability vectors) are considered and described below.

- Case 1: Set $\bar{\kappa}_j = 1/d + (1/6d) \cdot z_j$ with $z_j \overset{iid}{\sim} Bernoulli(1/2)$ for all $1 \le j \le d$, and normalize to get $\kappa_j = \bar{\kappa}_j/(\sum_j \bar{\kappa}_j)$. Set $\bar{\theta}_j = \kappa_j/a$ for $1 \le j \le d/2$, $\bar{\theta}_j = (2 - 1/a)\kappa_j$ for $d/2 + 1 \le j \le d$ with some $a > 1$, and normalize to get $\theta_j = \bar{\theta}_j/(\sum_j \bar{\theta}_j)$.

- Case 2: Set $\kappa_j = 2^{-d/2} \frac{(d/2-1)!}{(j-1)!(d/2-j)!}$ for $1 \le j \le d/2$, and $\kappa_j = 1/d$ for $d/2 + 1 \le j \le d$. Set $\theta_j = (2 - 1/a)\kappa_j$ for $1 \le j \le d/2$, and $\theta_j = \kappa_j/a$ for $d/2 + 1 \le j \le d$ with $a > 1$.

The positive number a above quantifies the deviation of $\kappa_{1:d}$ from the true $\theta_{1:d}$. It can be shown that $\bar{\rho}$ and μ in (46), and μ_{CS} in (50) are increasing in a when $a > 1$.

Table 10: *Type I Error Rates* $\times 10^2$ *at* $\alpha = 0.05$*, Case 2*

	Pearson's χ^2			Cross-sample χ^2_{CS}		Log likelihood ratio G			Hellinger H		
(d, n)	$\mathcal{N}(0,1)$	χ^2_{d-1}	$\chi^2_{V/2}$	$\mathcal{N}(0,1)$	χ^2_{d-1}	$\mathcal{N}(0,1)$	$\chi^2_{\mu_G}$	$\chi^2_{V_G/2}$	$\mathcal{N}(0,1)$	$\chi^2_{\mu_H}$	$\chi^2_{V_H/2}$
(40, 40)	0.16	6.48	0.16	**4.67**	3.73	5.88	2.55	**4.58**	5.64	4.28	**4.67**
(40, 100)	0.38	6.08	0.38	**4.92**	4.00	6.00	**4.99**	**4.88**	6.09	10.65	5.32
(40, 400)	1.24	5.38	1.18	**5.18**	4.27	6.16	**4.98**	**5.04**	6.16	6.16	5.18
(100, 40)	0.00	3.86	0.00	3.92	3.31	6.04	0.87	**4.80**	**5.93**	0.77	**4.84**
(100, 100)	0.00	3.61	0.00	3.79	3.23	5.64	2.40	**4.68**	**5.43**	4.07	**4.73**
(100, 400)	0.00	3.57	0.00	3.78	3.24	5.82	**5.43**	**5.03**	6.21	10.90	**5.65**
(400, 40)	0.00	1.97	2.66	3.57	3.23	**5.52**	0.60	**4.54**	**5.34**	0.15	4.33
(400, 100)	0.00	1.95	2.50	2.92	2.65	**5.77**	0.60	**4.94**	5.89	0.48	**5.13**
(400, 400)	0.00	1.69	2.49	2.58	2.34	5.33	2.09	**4.77**	**5.16**	3.71	**4.76**

We vary (d, n, a) for each type of κ, and compare the empirical and theoretical powers in every setting. Given a $(\kappa_{1:d}, d, n, a)$, the empirical powers are computed by repeating the following steps independently for 1×10^4 times: sample the data vector $(X_1, \ldots, X_d)$ from multinomial$(n, \theta_1, \ldots, \theta_d)$; compute the test statistics χ^2, χ^2_{CS} under the null probability vector $\kappa_{1:d}$; choose the cut-off values for χ^2 $-(d-1)$ and χ^2_{CS} at level α to be, respectively, $\Phi^{-1}_{V/2}(1-\alpha) - V/2$ and $\Phi^{-1}_{d-1}(1-\alpha) - (d-1)$ with $\Phi^{-1}_{\omega}(1-\alpha)$ being the $1-\alpha$ quantile of χ^2_ω distribution and V as in (33). The theoretical powers are computed as follows. Theorem 2.11 states that when $\kappa_{1:d} \neq \theta_{1:d}$, the variance of χ^2 approximates the variance of a non-central chi-squared distribution $\chi^2_\omega(\tau)$ with degrees of freedom ω and noncentrality parameter τ, that is

$$2\omega + 4\tau \approx V',$$

where V' is as in (47). We choose ω and τ as

$$\omega = V'/2, \quad \tau = 0.$$

The theoretical power of the Pearson's chi-square test is

$$\begin{aligned}
&\mathbb{P}\Big\{\chi^2_\omega(\tau) - (\omega+\tau) + \mu - (d-1) \geq \Phi^{-1}_{V/2}(1-\alpha) - V/2\Big\} \\
=&1 - \Phi_{\omega,\tau}\Big(\Phi^{-1}_{V/2}(1-\alpha) - V/2 + (d-1) - \mu + (\omega+\tau)\Big) \\
=&1 - \Phi_{\omega,\tau}\Big(\Phi^{-1}_{V/2}(1-\alpha) - V/2 + 3(d-1)/2 + V'/4 - \mu\Big),
\end{aligned}$$

where $\Phi_{\omega,\tau}(\cdot)$ is the CDF of the non-central chi-square distribution $\chi^2_\omega(\tau)$, and μ is as in (46). Similarly, by Theorem 2.12, the variance of χ^2_{CS} approximates the variance of a non-central chi-squared distribution $\chi^2_{\omega_{\text{CS}}}(\tau_{\text{CS}})$ with degrees of freedom ω_{CS} and noncentrality parameter τ_{CS} satisfying

$$2\omega_{\text{CS}} + 4\tau_{\text{CS}} = V_{\text{CS}},$$

where V_{CS} is as in (51). Picking ω_{CS} and τ_{CS} as

$$\omega_{\text{CS}} = V_{\text{CS}}/2, \quad \tau_{\text{CS}} = 0$$

the theoretical power of the cross-sample version test is

$$\begin{aligned}
&\mathbb{P}\left\{\chi^2_{\omega_{\mathrm{CS}}}(\tau_{\mathrm{CS}}) - (\omega_{\mathrm{CS}} + \tau_{\mathrm{CS}}) + \mu_{\mathrm{CS}} \geq \Phi^{-1}_{d-1}(1-\alpha) - (d-1)\right\} \\
=&1 - \Phi_{\omega_{\mathrm{CS}},\tau_{\mathrm{CS}}}\left(\Phi^{-1}_{d-1}(1-\alpha) - (d-1) + (\omega_{\mathrm{CS}} + \tau_{\mathrm{CS}}) - \mu_{\mathrm{CS}}\right) \\
=&1 - \Phi_{\omega_{\mathrm{CS}},\tau_{\mathrm{CS}}}\left(\Phi^{-1}_{d-1}(1-\alpha) + V_{\mathrm{CS}}/4 - (d-1)/2 - \mu_{\mathrm{CS}}\right)
\end{aligned}$$

with μ_{CS} as in (50). When $\theta_{1:d} = \kappa_{1:d}$, the theoretical powers for both tests are α.

As shown in Figure 3 and Figure 4, the cross-sample version statistic is more stable and robust than Pearson's chi-squared statistic. For the cross-sample version, the theoretical powers (orange) predict accurately the empirical powers (black) across all settings. While Pearson's chi-squared statistic fails when the cell probabilities are unbalanced.

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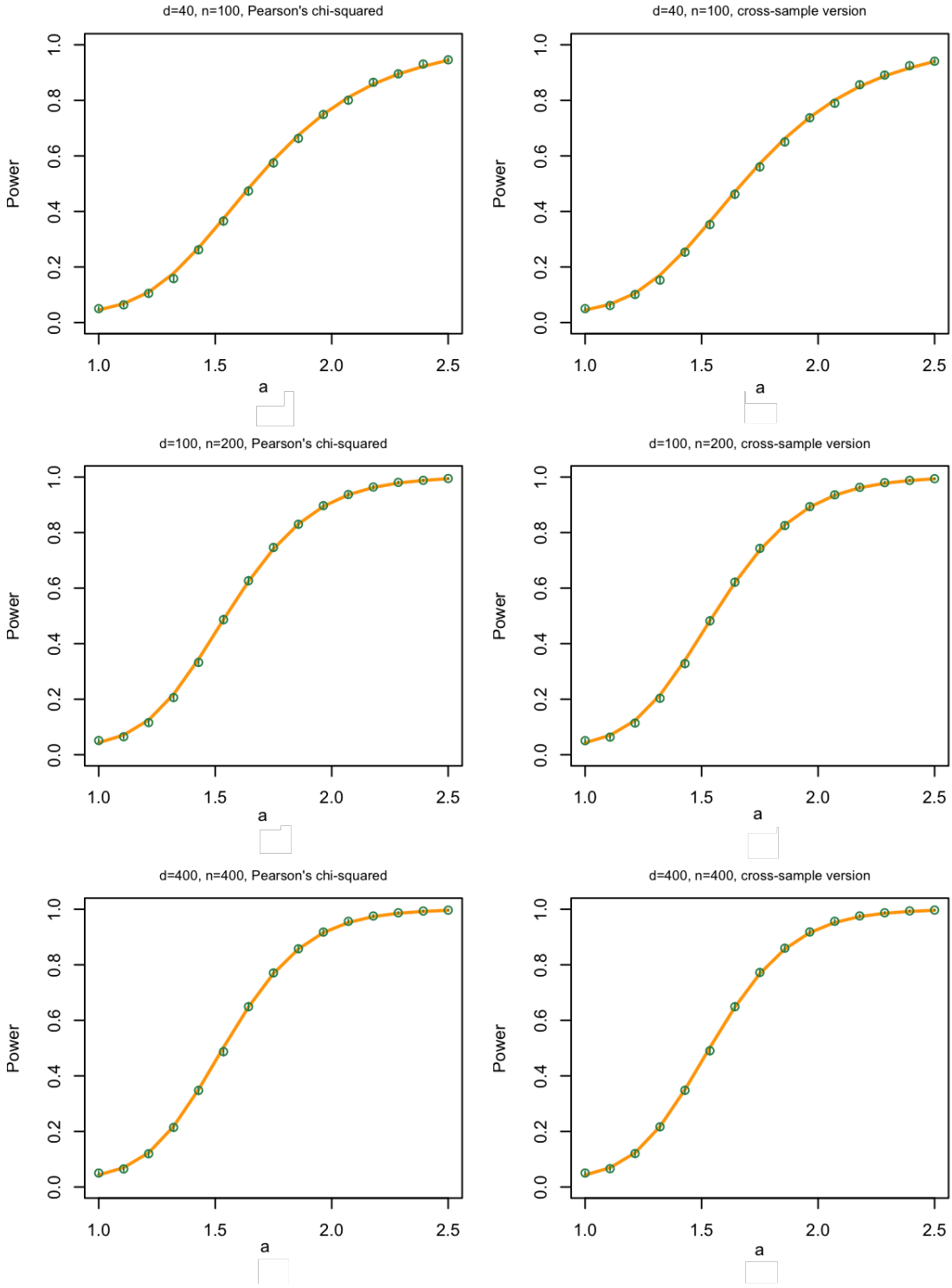


Figure 3: The orange solid line depicts the change of theoretical power with the deviation from the null probability vector. The black circles show the scatterings of the empirical powers with the error bars representing minus and plus three standard deviation.

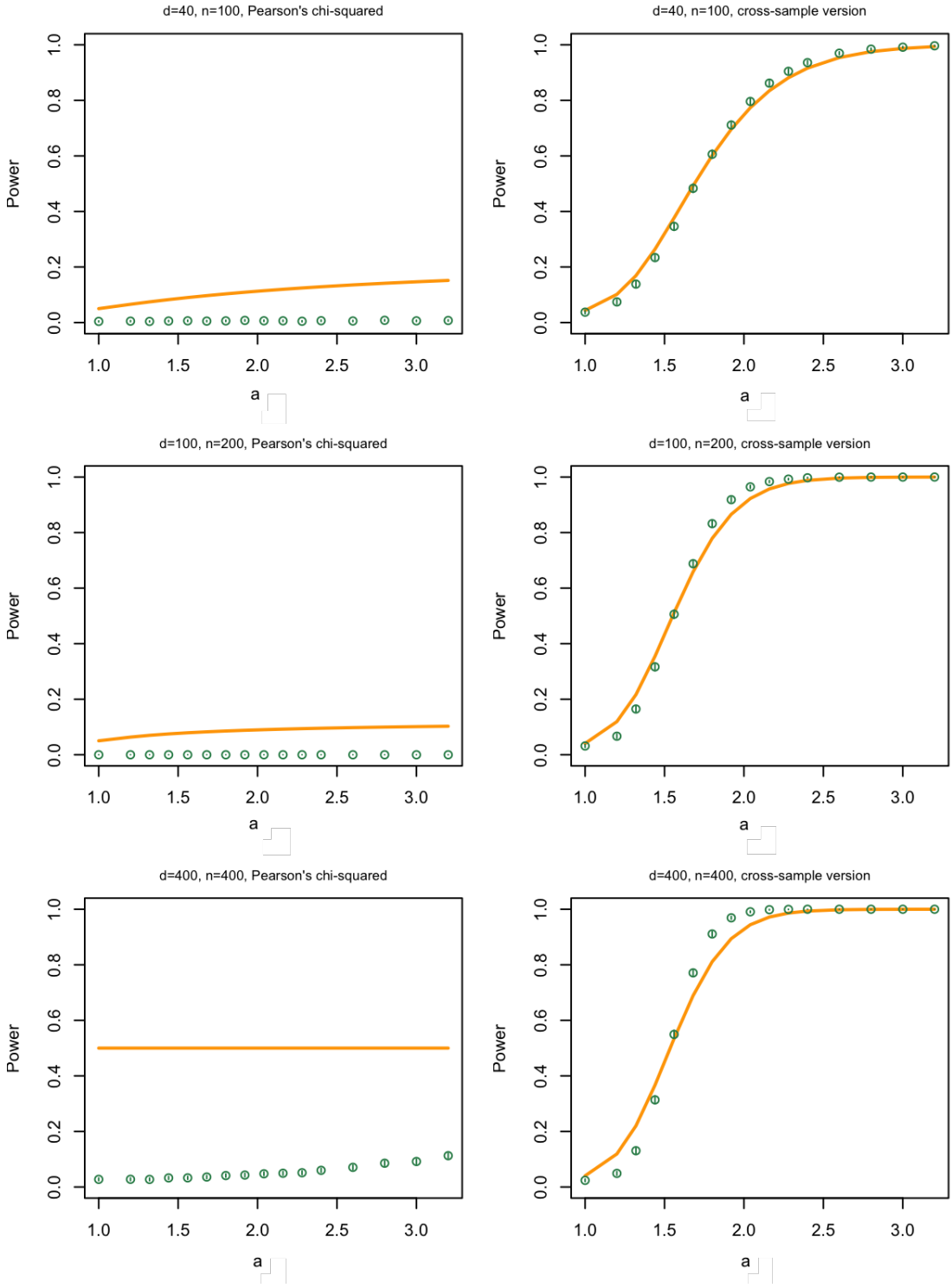


Figure 4: The change of theoretical powers along the deviation from the null is depicted by the orange solid line. The black circles show the scattering of empirical powers with the error bar representing minus and plus three standard deviation.

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